Survey Research 2020-05-06

2020 K-Innovation ODA Program with Cambodia

Capacity Building and Pilot Implementation Program on Technology Roadmap

SCIENCE AND TECHNOLOGY POLICY INSTITUTE

2020 K-Innovation ODA Program with Cambodia

Title	Capacity Building and Pilot Implementation Program on Technology Roadmap
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Partner Country	The Kingdom of Cambodia
In Cooperation with	Cambodia National Science and Technology Council (NSTC)
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Executive Summary

Recently, Cambodia's science and technology sector has undergone major changes. The STI Policy (2019-2025), which was established as a result of the K-Innovation project in 2018, was revised to STI Policy (2020-2030) and finally approved by the government at the end of last year. Cambodia's first science and technology innovation plan has been established. Also, in early 2020, there was a major change in governance. The Ministry of Industry and Handcraft was changed to the Ministry of Industry, Science, Technology and Innovation (MISTI), and as a sub-organization, the General Department of Science, Technology and Innovation, and the National Institute of Science and Innovation were established. In the ongoing "Rectangular Strategy Ⅳ 2019-2023," detailed strategies include "human resource development in the science and technology sector" and "diversification of Cambodia's economic structure through energy and digital technologies and fourth industrial revolution". These are based on science and technology, and recent governance and policy changes in Cambodia can be seen as an expression of its willingness to realize them.

The project is designed as following project of the project("Policy Consultation on National STI Policy Development and STI System Management for Cambodia's Policymakers") from 2018 to 2019. Through the first phase of the project, STEPI had provided consultations for establishing Cambodia's STI policy, and Cambodian policymakers recognized the importance of technology roadmap and wrote pilot roadmap through sharing experiences and training in Korea. The second phase of the project, which will take place over the next three years, aims to develop Cambodia's strategic technologies for successful implementation of key STI policies, including STI Policy 2020-2030 and to establish national technology strategies and technology roadmaps that can actually be applied to the policy.

Like the first phase of the project, it was expected to visit each other's countries to interact and conduct on-site surveys, but the plan was revised completely due to COVID-19, which has been sweeping the world. STEPI and GS-NSTC decided to make a service contract to select Cambodia's national strategic technologies by GS-NSTC and STI experts in Cambodia with advice from Korean experts. All discussions were conducted in writing and video conferences, and researchers from both countries made utmost efforts to minimize the limitations of communication and the negative impact on the project.

The consultation process for the project had the following activities:

With the recognition on the seriousness of the COVID-19 outbreak, STEPI suggested a service contract for survey and operation of expert committees conducted by GS-NSTC. Detailed discussions for service implementation were conducted by e-mail and video conference.

The contract was signed in June 2020. Cambodia's main tasks were research and analysis on Cambodia’s STI policy and environment, establishment of expert committees, selection of Cambodia's six strategic technology groups through surveys and consultation, and preparation of reports. Korean researchers prepared and provided survey guidelines and list of candidate technology groups for expert surveys.

A video conference for interim review was held in July 2020. Korean researchers reviewed the survey questions shared by GS-NSTC and provided some comments, while GS-NSTC held the first meeting of a 17-member expert committee in August.

The report was finalized and submitted in late October after complementary process of first draft of the report with Korean researchers. The main results of the survey, which involved a total of 50 experts, are as follows. Cambodia's strategic technology group was followed by ICT (60%), agricultural technology (14%), and biohealth technology (12%), and computer software, food emanation products, veterinary medicine, food microbiology, and nutrition.

This year, there were many difficulties due to the difficulties of COVID-19, but it was able to achieve more than expected. First, Cambodia's national strategic technology was derived as a necessary procedure for establishing a national strategic technology roadmap. ICT, agricultural technology, and biohealth will be selected as the top strategic technologies and reflected in the technology roadmap of six detailed strategic technologies to be implemented from 2021.

In addition, both Korea and Cambodia have accumulated experience in dealing with unpredictable factors that may arise in ODA projects, such as COVID-19. In 2021, it is expected that direct exchanges due to COVID-19 will be difficult, and we expect to make preemptive business plans based on this year's experience.

Finally, the effectiveness of the project was confirmed in enhancing Cambodia's capabilities. GS-NSTC carried out the process of deriving national strategic technologies in the event of a sudden disaster. The project was organized so that exchanges and education and training conducted through the first phase of the project could be connected organically to the second phase of the project, and both Korean and Cambodian researchers participated in the second phase of the project without much change.

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Project Overview

Capacity Building and Pilot Implementation
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Chapter 1. Project Overview

1. Introduction

The Royal Government of Cambodia is striving to achieve its Rectangular Strategy Phase IV through the 2019-2023 National Strategic Development Plan (NSDP) with the mid- and long-term vision of becoming an Upper Middle-Income Country by 2030 and a High-Income Country by 2050.

Driven by its long-term vision toward 2030, particularly as expressed in the Rectangular Strategies III and IV, Cambodia has prepared and adopted the Cambodian Industrial Development Policy 2015-2025 (IDP 2015-2025) as a guide to promoting the country's industrial development, which will help maintain sustainable, inclusive high economic growth and reach Middle-Income economy status.

In this regard, Cambodia has promoted science, technology, and innovation (STI) development in order to achieve sustainable socioeconomic development goals. Recently, as part of this effort, the Cambodian government finally approved in January 2020 a bill to change the existing Ministry of Industry and Handicraft to the Ministry of Industry, Science, Technology, and Innovation (MISTI). Such change shows the Cambodian government's willingness to use science and technology as a springboard for Cambodia's economic development in line with the rapid pace of world science and technology development. In March 2020, the Law on the Establishment of the Ministry of Industry, Science, Technology, and Innovation was published, and the sub-decree on organization and function was revised. As a result, the General Department of Science, Technology, and Innovation and the National Institute of Science, Technology, and Innovation were established as sub-organizations of MISTI. This change of government is expected to raise further the government's interest in Cambodia's technological development.

The General Secretariat of the National Science and Technology Council (GS-NSTC), a partner in the project, is the first science and technology-related government agency established in Cambodia. GS-NSTC is responsible for the establishment and implementation of national STI policies and plans for Cambodia's socio-economic development. Its role is very important as an institution that manages the overall STI in Cambodia, from training human resources in science and technology and implementing national R&D projects in the science and technology field to establishing various laws and infrastructures such as STI knowledge sharing system.

In this regard, GS-NSTC aims to address various national challenges through cooperation with various countries. The K-Innovation project was also part of such cooperation, initiated at the request of the Cambodian government. Based on the confirmation and acceptance of the project proposal submitted by GS-NSTC, the K-Innovation ODA Program has affirmed the commitment to supporting Cambodia in making faster progress toward achieving its development goals.

At the same time, having recognized Cambodia as a major partner recently, the Korean government is actively promoting economic cooperation and economic development.

The aim of the project is to provide consultation on capacity building and pilot implementation program on the technology roadmap. This project is a follow-up to the capacity building for national STI policy development and STI system management for Cambridge policymakers, conducted between 2018 and 2019. It includes the following aspects:

1) Delivery of survey design and management methodologies and guidelines

2) Selection of national strategic technologies that meet the current status of Cambodia

3) Sharing of the Korean experience relevant to Cambodia

As a government-affiliated policy think tank, STEPI has been conducting policy research for Korea's science and technology innovation for about 60 years. STEPI has close cooperative relationships with various related ministries, local governments, academe, and industries. As the role of science and technology innovation has become important in the social and economic development of developing countries, the K-Innovation project has established cooperative relations with various countries to share Korea's experience and provide policy advice and education & training for STI development.

As an output of the 2020 K-Innovation Program in Cambodia, GS-NSTC established a final draft of the “Cambodian STI Policy 2019-2025” with support and policy consultation from STEPI. In addition, GS-NSTC and STEPI agreed that close and strategic linkage between the national economic development plan and science and technology development is necessary for the effective implementation of the STI policy. Strategic priority should also be set among the STI areas in STI policy for efficient R&D investment. In other words, for the implementation of the STI policy, Cambodia needs a national technology strategy as a next step.

2. Objectives

The title of the project is "Capacity Building and Pilot Implementation Program on the Technology Roadmap." The program runs for about 12 months, from January 1 to December 31, 2020.

The following are the objectives of the 2020 K-Innovation ODA Program with Cambodia:

- Delivering methodologies and establishing guidelines for national technology strategies and technology roadmaps
- Strengthening the capacity of Cambodian policymakers and researchers to select a national strategic technology
- Selection of Cambodia's national strategic technology (pilot)

3. Project Framework

■ Consultation on the selection of national strategic technologies

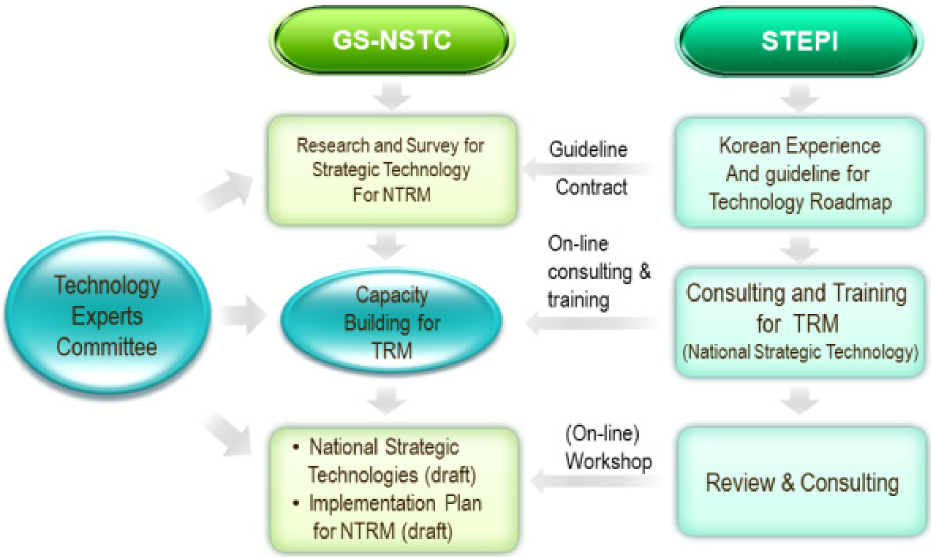
The consultation process for choosing the national strategic technologies will consist of the following activities:

- Cambodia's domestic experts and STEPI researchers conduct research and diagnosis.
- The STEPI research team provides survey guides for expert surveys and interviews and conducts online consulting on questionnaires and survey analysis reports written in Cambodia.
- The Cambodian research teams create expert committees and conduct surveys and interviews.

■ Strengthening the STI strategy building capacity of Cambodia with public-private partnership

A training for selecting the strategic technology of Cambodia will be held in writing and via videoconference. With the survey and data analysis conducted by Cambodian researchers as the main activities of this research, GS-NSTC reports about each stage of the research, and STEPI provides feedback on it. Through this process, GS-NSTC is expected to strengthen the capacity to select Cambodia’s strategic technology and mobilize STI experts in the academe, industries, and other related ministries.

[Figure 1-1] Conceptual Framework of the Project



4. Project Team

4.1 Research Team

A research team was formed considering the research area and goal above, and a high-level adviser was appointed in accordance with the relevant expertise in the fields of science and technology policy, related knowledge, and know-how for sharing the Korean development experience.

The research team is composed of full-time experts including Dr. Jungwon Lee as Project Manager (PM) and Ms. Jieun Kim as a full-time researcher. They are responsible for technical advice on selecting Cambodian strategic technology by investigating the policies of both countries and providing guidelines for research. Especially this year, it is important to coordinate the project to response to the changes in research plan due to external influences, such as global disasters, in partnership with GS-NSTC. An external expert, Dr. Hyun Yim has been identified for the delivery of consultation on the process of strategic technology selection based on the topics to be covered. He participated in the Cambodia project as a lecturer of the TRM capacity-building workshop in 2019. By participating in this year's research, Cambodian researchers' learning of the TRM theory can lead to practice in selecting national strategic technologies more systemically.

[Table 1-1] Research Team

Name	Organization	Position & Area of Expertise	Contact Information
Jungwon Lee	STEPI, Senior Research Fellow Project Manager (PM)	Innovation Ecosystem	leejw@stepi.re.kr
Hyun Yim	KISTEP, Senior Research Fellow	TRM Practice	hyim@kistep.re.kr
Jieun Kim	STEPI, Researcher	Project Coordinator	kjieun@stepi.re.kr

4.2 Local Research Team

The research members of GS-NSTC are responsible for selecting the strategic technology of Cambodia, including design, survey, and analysis of questionnaire and data. The local research team also convenes an experts' committee to collect advice reflecting Cambodia's political, economic, and technical status. The GS-NSTC team works in cooperation with the STEPI research team.

[Table 1-2] Local Research Team

Name	Office	Position	Contact Information
Teav Rongsa	GS-NSTC	Secretary General (Acting)	teavrongsa@gmail.com
Ung Viseth	GS-NSTC	Deputy Secretary General	viseth_ung@yahoo.com
Try Rithea	GS-NSTC	Deputy Secretary General	-

5. Project's Main Activities

5.1 COVID-19 Outbreak

At the end of the year 2019, an unidentified virus later called COVID-19 broke out and started to threaten the entire world. In order to prevent the spread of the disease, governments have begun restricting the number of people entering foreign countries. Cambodia has also restricted all foreigners from entering the country since March 30. Similarly, Korea has begun imposing restrictions on those who enter the country for two weeks, including mandatory self-practice measures. In fact, researchers from South Korea and Cambodia have discussed ways to overcome the situation as it has become impossible to proceed with the project through travel.

5.2 Activity 1: Preliminary Meeting

- The COVID-19 situation in both countries was identified through e-mail, and how to operate the project was discussed.
- The first workshop in Cambodia and capacity building in Korea, which had been scheduled for the first half of the year, were judged to be difficult due to the worsening of the COVID-19 situation between the two countries.
- STEPI and GS-NSTC have sought ways to proceed without direct interaction. STEPI, led by GS-NSTC, decided to propose a service contract to conduct expert surveys, convene expert committees, and hold videoconferences to discuss details.

5.3 Activity 2: Videoconference

- On April 22, 2020, the first videoconference was held to coordinate business operations. Dr. Jungwon Lee and Ms. Jiun Kim represented STEPI, with H.E. Has Bunton, H.E. Viseth Ung, and Mr. Nhem Solivan representing GS-NSTC.

5.4 Activity 3: Conclusion of Service Contract

- As a result of exchanging and coordinating opinions between STEPI and GS-NSTC on the service contract, including the period of service performance and scope of its role, the service contract between the two agencies was concluded in June 2020.
- The service lasted about five months until November, and GS-NSTC conducted a survey to select the strategic technology of Cambodia; based on this, it produced a report and submitted it to STEPI.

5.5 Activity 4: Selection of Strategic Technologies

- The main purpose of the contract is to select Cambodia's national strategic technology. To this end, Cambodia's science and technology policy and STI environment was first investigated, followed by the establishment of Cambodia's expert committee and selection of Cambodia's six strategic technical forces through surveys and interviews. The entire process was led by GS-NSTC, with the STEPI team providing consulting in all processes, including guidelines for filling out questionnaires.

6. Project Schedule

The overall project schedule is as follows:

[Table 1-3] Project Schedule

Activities	Jan. '20	Feb. '20	Mar .'20	Apr. '20	May '20	Jun. '20	Jul. '20	Aug. '20	Sep. '20	Oct. '20	Nov. '20	Dec. '20
Project Identification												
Plan Adjustment												
Conclusion of Service Contract												
Survey Design												
Sample Selection and Questionnaire Design												
Expert Committee Meeting												
Conduct of Survey												
Data Analysis												
Final Report												

7. Project Outputs

STEPI provided the following deliverables:

- Deliverable 1: Project Proposal (April 2020)
- Deliverable 2: Guidelines for Survey Design (June 2020)
- Deliverable 3: List of Candidate Technologies (June 2020)
- Deliverable 4: Interim Report (July 2020)
- Deliverable 5: Final Report (December 2020)

GS-NSTC provided the following deliverables:

- Deliverable 1: Survey Result Report (October 2020)
- Deliverable 2: Raw Data of the Survey (November 2020)

CHAPTER 2

Capacity Development Process

Capacity Building and Pilot Implementation Program on Technology Roadmap

Chapter 2. Capacity Development Process

1. Meetings for Contract and Implementation of Services

1.1 Summary of Meetings

The preliminary meeting between the two agencies was held through several e-mail exchanges. As the COVID-19 situation in Korea and Cambodia worsened, and restrictions on foreigners' entry and exit were tightened, the first workshop scheduled for the first half of the year and the capacity-building workshop in Korea were found to be difficult. Thus, the two agencies exchanged ideas for a service contract wherein GS-NSTC conducts a local expert survey and convenes an expert committee so that the project can proceed without a direct visit. Since this type of service contract was the first of its kind in the K-Innovation project, videoconferencing was planned to identify the regulations of each agency for the conclusion of the contract and discuss the details and conditions of the service, including the scope of the service.

The first videoconference was held on April 22, 2020. We shared the domestic status of Korea and Cambodia. As COVID-19 is expected to persist worldwide, all planned activities for the first half will likely be non-face-to-face, and the schedule for the second half will also need to be adjusted as necessary. At this time, too, the Cambodian government was implementing a reorganization, which was highly likely to cause changes in the overall Cambodian government.

STEPI requested data on Cambodia's experience in developing science and technology policies and strategies for carrying out projects in 2020. It also called for the establishment of an expert committee to derive Cambodia's national strategic technologies and approve the “Technology Roadmap Implementation Plan 2021-2022.” The two agencies agreed to set up a panel of experts from the government, academe, and industry consisting of about 10 people.

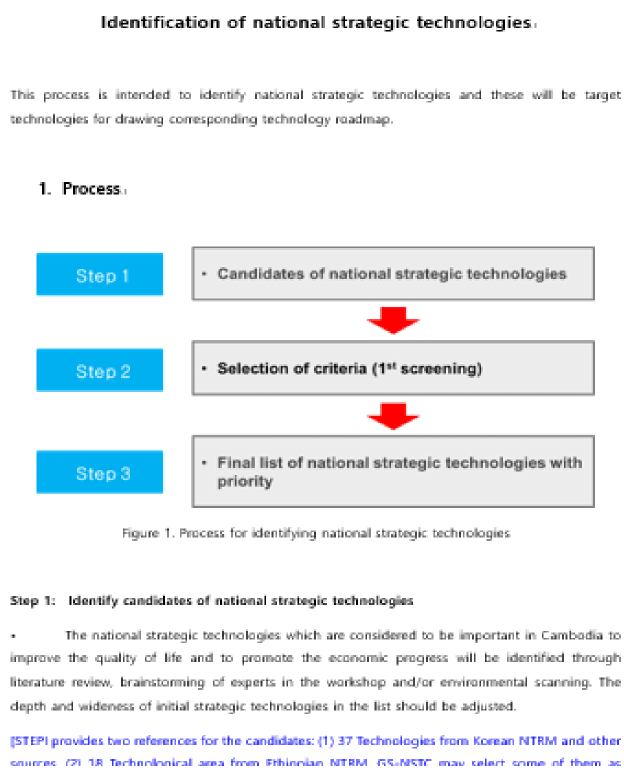
After the service contract was concluded, the second videoconference was held on July 8, 2020. Shortly after the conclusion of the service contract, the Secretary General, H.E. Has Bunton, and Mr. Neth Vansitha moved to another agency. H.E. Teav Rongsa was appointed as acting secretary-general; fortunately, he recognized the importance of this project and proceeded with the project as originally contracted. Accordingly, GS-NSTC decided to form an expert committee of around 10 people in July and share it with STEPI. GS-NSTC will complete the questionnaire in accordance with the guidelines provided by STEP and hold its first expert committee meeting in August. Through the survey, six national strategic technologies will be selected: three in 2021 and three in 2022.

2. National Strategic Technology Selection

2.1 Survey Design

Since the signing of the service contract, Korean researchers, including STEPI researchers, have provided GS-NSTC with survey guidelines and lists of candidate technical groups for the expert surveys to be conducted in Cambodia. The survey guidelines presented the 3rd step for selecting strategic technologies as 1) national strategic technical candidates, 2) primary screening considering the economic ripple effect and practicability, and 3) final technical group selection. It also included a questionnaire sample. The list of candidate technology groups consisted of technologies from 37 countries, including Korea's National Science and Technology Roadmap, and 18 countries by reviewing data from Ethiopia's National Science and Technology Roadmap.

[Figure 2-1] Survey Guideline



Identification of national strategic technologies

This process is intended to identify national strategic technologies and these will be target technologies for drawing corresponding technology roadmap.

1. Process

Figure 1. Process for identifying national strategic technologies

Step 1: Identify candidates of national strategic technologies

• The national strategic technologies which are considered to be important in Cambodia to improve the quality of life and to promote the economic progress will be identified through literature review, brainstorming of experts in the workshop and/or environmental scanning. The depth and wideness of initial strategic technologies in the list should be adjusted.

[STEPI provides two references for the candidates: (1) 37 Technologies from Korean NTRM and other sources, (2) 18 Technological area from Ethiopian NTRM. GS-NSTC may select some of them as

[Figure 2-2] Candidate Technologies for the National Strategic Technology of Cambodia

Candidates of technology for the National Strategic Technology of Cambodia

Area	Product/Technology
ICT	Optical communication technology
	High speed mobile communication technology (5G)
	Mobile multimedia contents
	Intelligent network
	Wearable computer
	Digital broadcasting
	E-commerce
	Fintech
	Information security
	Intelligent robot
	Aritificial intelligence
	Home networking
	Intelligent medical system
Bio & Helath	New Drug discovery and development (chemical)
	New Drug discovery and development (natural)
	Diesease diagnostics, treatment, and prevention
Environment & Energy	Recution of environmental pollution
	Recycleing system
	Natural disaster prediction and damage reduction
	Engergy saving
	Renewable energy
	Fuel cell
Manufacturing	Transpotation vehicles
	Road and building construction
	Smart Factory
	Machinery
	New material development (metal, ceramic, polymer)
Agriculture	Agricultural machinery
	Seeds development
	Fertilizer
	Agricultural water management
	Agricultural products processing
	Digitalization of agriculture
National prestige & security	Defense technology
	Aerospace technology
	Food security
	Mekong River Development

2.2 Convening Expert Committees and Conducting Surveys

Based on the guidelines, GS-NSTC researchers prepared survey questionnaires and shared the first draft on July 29, 2020 and the second draft on August 5, 2020. Korean researchers reviewed it and delivered comments to improve it. It was suggested that selecting all the candidate technologies as strategic technology is impossible because the expertise of each expert is different. Thus, it was modified so that all strategic skills of one candidate technology unit in which each expert has expertise are selected.

GS-NSTC formed a 17-member panel of experts to hold its first meeting in early August. The panel of experts is composed of government (GS-NSTC, Ministry of Health, Ministry of Industry, Science, Technology, and Innovation, Ministry of Post and Telecommunications, etc.) and research institutes (Cambodia Institute of Technology, Cambodia Development Research Institute, AAEOMman, and Industry Research Institute (AEOC)). As a result of the first meeting, experts suggested to GS-NSTC that a more in-depth investigation of Cambodia's strategic technologies is needed, including discussions with Korean researchers on each technology group, in addition to the survey.

[Figure 2-3] Cambodia’s Local Expert Committee

3. Sharing and Consulting on Survey Results[1]

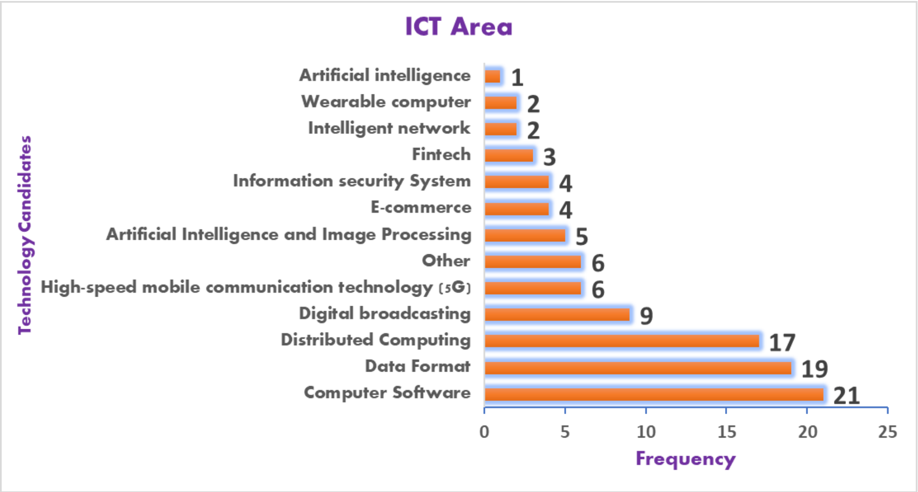
3.1 Brief Introduction of Survey Result

The scale of the respondents is 50, and they are all experts with extensive knowledge of science and technology and field experience. In order to obtain a more accurate and valid answer, respondents were organized mainly in the senior positions of their agencies. Of the 50 respondents, 44 were male and 6 were female; those in their 30s constituted the biggest portion with 70 percent, followed by those in their 20s with 18 percent and those in their 40s with 10 percent. There were 17 bachelor's degree holders, 17 master's degree holders, and 16 PhD holders. The respondents' answers were directly obtained by GS-NSTC officials as they could identify the current status and limitations of Cambodia's technology development. Respondents selected 10 detailed candidate strategic technologies for each candidate group based on policy relevance, feasibility, and importance.

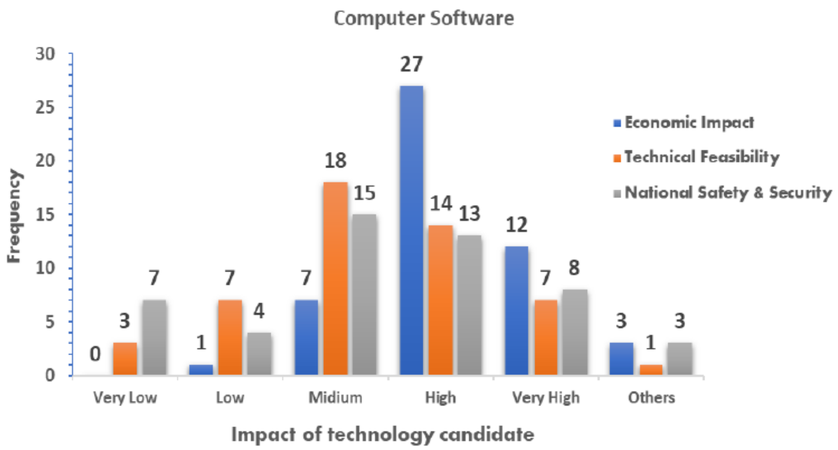
According to the survey, 60% of respondents chose ICT technology as the most important strategic technology, followed by agricultural technology with 14% and biohealth technology with 12%. When asked about detailed strategic technologies that should be developed first within the ICT, agricultural, and bio-health technology groups, 24% of respondents cited computer software. Food emulsion products (18 percent), veterinary medicine (14 percent), food microbiology (14 percent), and nutrition (12 percent) followed with similar rates. Interviews and surveys show that computer software, which is also the most important in the ICT sector, has a significant impact on Cambodia's economy and on national safety and security.

[1] The detailed findings are accompanied by a report prepared by GS-NSTC.

[Figure 2-4] Strategic Technology: ICT Area



[Figure 2-5] Evaluation of Economic Impact, Feasibility, and National Security Aspect of Computer Software Technology



CHAPTER 3

STI Policies of Cambodia

Capacity Building and Pilot Implementation Program on Technology Roadmap

Chapter 3. STI Policies of Cambodia

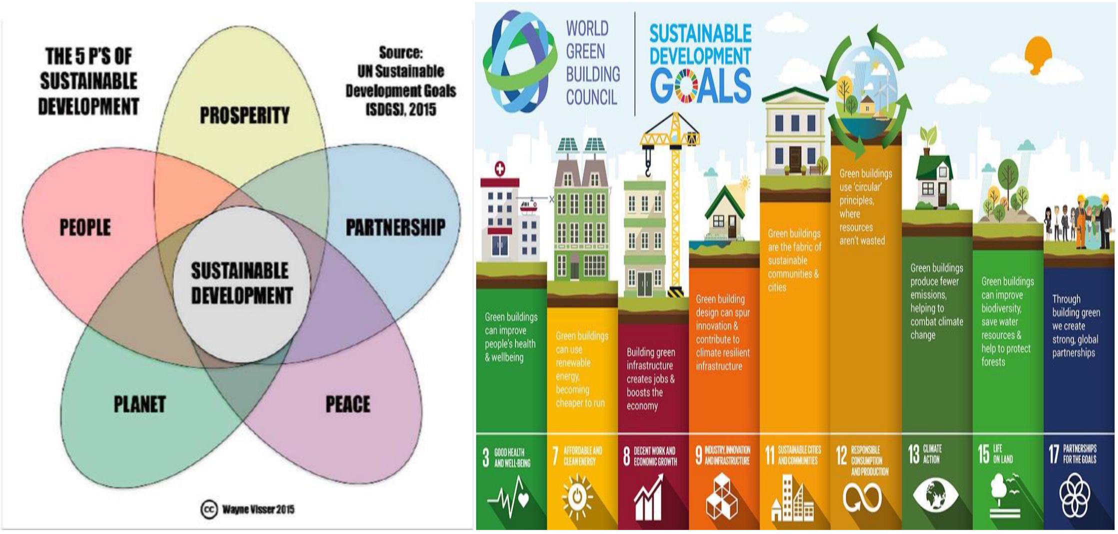
1. Cambodia's Sustainable Development Goals (2016-2030)

1.1 Introduction

The SDGs emphasize the wider Sustainable Development Agenda adopted by the United Nations General Assembly's 2015 session and herald new priorities and thoughts on global development needs and directions. Remarkably, the SDGs classify the identical global priorities of delivering environmental sustainability (and precisely the threat posed by climate change) and the ongoing priority of eliminating poverty. They are likewise more spread out, numbering 17 goals and 169 targets (compared to the MDG's 8 goals and 48 targets), and they are evidently more motivated by looking for complete reductions and strict equalities, whereas the previous Millennium Development Goals (MDGs) had 8 goals and 48 targets.

Meaningfully, the SDGs' intention is to be transformative toward sustainable development for all nations, contributing both a guide to policy sets and a means of inspection of national progress. They obviously know the importance of equity within and between states, the universality of human needs and rights, and the interconnectedness of the development challenge. The 17 goals are clustered under 5 themes, the so-called 5Ps—prosperity, people, planet, peace, and partnership—that represent a networked whole (see Figure 3-1).

[Figure 3-1] Global Goals



The SDGs build on, but are typically different from, the forerunner MDGs. The fresh goals apply to all individuals and all states so that no one—people, country, or group—is left behind. The approach declares that the task of delivery is everyone's business—the state, public and private sectors, and citizens themselves—and that the SDGs are owned and driven by each nation based on its own priorities, resources, and capacities. By itself, they are not an access point for the transfer of new streams of foreign aid; rather, their accomplishment rests for the most part on national leadership and resources. Each UN member state is called on to realize its commitments by nationalizing the goals to fund their own development plan and to enable effective progress monitoring and evaluation. States are free to adapt the global framework as they see fit with the nation's context, drawing on the global goals, targets, and indicators.

The process was started in the Kingdom in late 2015 by studying and mapping the global goals and targets to national priorities. A total of 17 SDGs were selected, and 1 extra goal related to the clearance of land mines and Explosive Remnants of War (reflecting the national priority of de-mining of the Nation's territory) was added. This resulted in a final version consisting of the following: 18 Cambodian Sustainable Development Goals, 88 nationally important targets, and 148 (global and locally defined) indicators 96 of which are national indicators. This monitoring framework for CSDG is reviewed in Chapter III (and detailed in Schedules 1 and 2 in Part 2). The Royal Government of Cambodia has yet to approve officially the embarking on a VNR but envisages that this will have an effect during 2019.

The CSDG framework, together with the Kingdom's socioeconomic platform set forth in the Rectangular Strategy (RS) will offer the basis for the novel series of the National Strategic Development Plan (NSDP). The framework presented here will be subject to further consultations and eventual adoption by the RGC, which will make the new Rectangular Strategy certified after its succeeding election, which will take place in July 2018. It is also good to keep in mind that, compared to the global goals, the CSDGs have a longer period (up to 2030), so they serve as a counterpart to the Government's Vision 2050.

The CSDGs likewise place great weight on leaving no one behind so that all Cambodians share in the state's forthcoming development and prosperity. This is matched with a guarantee of sustainability as well as developing while also protecting the nation's abundant natural capital for current and future generations and playing the Kingdom's part in opposing climate change.

1.2 Objectives

The CSDG Framework has four precise objectives:

- Presentation of the national goals, targets, and indicators based on the Kingdom's priorities;
- Identification of the organizations responsible for oversight and leading of activities to achieve the targets and monitoring schedules;
- Identification of data sources for each indicator and the data cycle, with a provision of the working definitions and methods for computing indicators;
- Presentation of pathways toward the accomplishment of targets, setting (2015) of national baselines, setting of annual (or cycle-based) target values, and implementation at the sub-national level and regional level (CSDG, 2016).

2. National Strategic Development Plan and Rectangular Strategy

2.1 National Strategic Development Plan (2014-2018)

The Royal Government of Cambodia (RGC) had been developing the National Strategic Development Plan (NSDP), which was the backbone of these efforts, and it is focuses on overcoming binding constraints, bringing about the settings for the expansion and excavation of the economy, safeguarding progressive development in socio-economic conditions, mobilizing resources, and monitoring progress.

The NSDP in turn was characterized by two further strategic processes: first, by the Rectangular Strategy (RS), which expresses the socio-economic scopes of the government's political platform for the existing parliamentary term; second and more lately, by the state's long-term visioning—Vision 2030, which sets a route to graduation from the group of low- to middle-income states and to upper middle-income status, and Vision 2050, which targets the attainment of high income status.

For NSDP 2019-2023, the pace and focus will change all over again to adapting the economic model to the new challenges posed by the quickness of Cambodia's regional and global integration, falling ODA, and likely loss of trade privileges. With 2015 as the base year for the CSDGs, NSDP will also frame the establishment of the new delivery capacities and approaches required by the CSDGs agenda and address early priorities, mainly the completion of any unfinished business from the CMDGs.

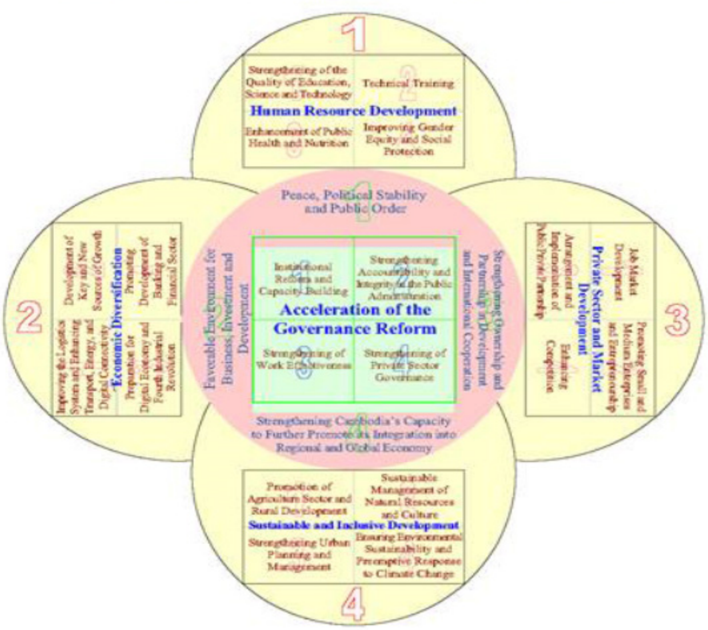
Additionally, the Plan will consist of seven themes that will emerge from the Mid-Term Review in 2016 of the existing cycle. These are: (1) stimulating poverty reduction and inclusive growth; (2) intensifying agriculture; (3) securing better competitiveness; (4) dealing with migration and urbanization; (5) fighting climate change and deforestation; (6) pursuing good governance; and (7) civilizing the human resource base.

2.2 Rectangular Strategy

2.2.1 Participants’ Comments on the Capacity Development Workshop

The RGC’s political platform for the 6th national assembly mandate and the Rectangular Strategy (RS) Phase IV will mutually serve as a comprehensive policy framework for formulating NSDP 2019-2023, which is articulated in the newly elected governments, and NSDP consequently symbolizes the primary mechanism for delivering the RS commitments. The development of RS IV will be completed after the national election in July 2018.

Figure. Schematic of the Rectangular Strategy (RS-Ⅳ)



Source: Royal Government of Cambodia (2018)

The RS offers successive iterations in four priority outcomes: Growth, Employment, Equity, and Efficiency. The recognized approach is articulated within 4 policy rectangles covering improved agriculture, lengthened physical infrastructure, private sector development and employment, and human resource development, with each consisting of 4 additional policy components. These are placed on a central good governance rectangle and are in turn showed by an analysis of the external environment.

Each phase of the RS has set an evolving set of policy objectives. For RS III, these were:

- Guaranteeing a high level of annual economic growth (above 7%) that is sustainable, inclusive, and equitable and resilient to astonishments, accomplished by diversification and enhanced competitiveness, and which maintains macroeconomic firmness.
- Making more higher-value, jobs, particularly for early people via the promotion of domestic and foreign investment. Attaining (at least) a 1 percentage point decrease in the poverty rate annually and appreciating Cambodia's Millennium Development Goals (CMDGs), while prioritizing human resource development and sustainable use of natural resources.
- Enlightening institutional capacity and governance at the national and sub-national levels and certifying the effectiveness of public services.

In introducing RS-III, Samdech Akka Moha Sena Padei Techo HUN SEN, Prime Minister of the Royal Government of Cambodia, said thus: "The government recognizes that it is essential to ensure consistency in terms of hierarchy, role, substance, coherence, and synchronization in the middle of the Rectangular Strategy, National Strategic Development Plan, sectoral development strategies, and other policy documents." Furthermore, he disclosed that the government's "strategy will be executed mostly through the National Strategic Development Plan" underlining the centrality of the NSDP and, in the context of the CSDGs, its importance in connecting the delivery of the long-term objectives to the policies and practices of the government.

The fourth phase of the Rectangular Strategy (RS-IV) will resolve the new challenges faced including the pressures as the Kingdom develops as an emerging middle-income country. The Royal Government identifies the obligation for strong focus on productivity and competitiveness (predominantly of labor). The significance of social development, as an adjunct to economic development—in areas like the labor market and social protection—

is also obligatory (RGC, 2018). This is in keeping with renewed commitments to the effective management of natural resources and preservation of Cambodia's environmental endowment, including responding to climate change.

3. Cambodia's Industrial Development Plan (2015-2025)

3.1 Overview of the Industrial Structure

Cambodia's industry remains weak with its modest structure, narrow base, and low level of sophistication; this is particularly true for the garment and food processing industries. Almost all production activities are family-based with the absence of entrepreneurship and insufficient use of technology; thus restraining their ability to compete in international markets. Its crucial characteristics can be summarized as follows: weak industrial base; misplaced middle and informal industrial structure; weak, urban-centered entrepreneurship; low value addition; and low level of technology application.

3.1.1 Narrow Industrial Base

In accordance with GDP, the industrial sector consists of three important activities: garment production; construction; and food and beverage processing. Other sectors have yet to make substantial inroads, with wood and paper processing and publishing even seeing a decline. The garment sector plays a vital role in the industrial sector with its share soaring from a mere 8.2% in 1993 to 51.8% in 2004 as the year of its peak growth. Later, it recorded a small decline to 42.4% in 2013. The construction sector, which accounts for 30% share of the industrial sector in 1993, has considerably declined to its lowest level of 20.2% in 1998 but has since bounced back to 30.1% in 2013. Food processing has been on a consistent decline from 32.7% in 1993 to about 10% during the last 5 years.

Almost all of the state's enterprises are in retail and food. Out of approximately 510,000 enterprises, only 70,000 or 14% are in manufacturing, 45% of which are in food and beverage processing and the remaining 35% are garment and textile enterprises. It should be noted that 80% of large industrial enterprises are in garment, textile, and footwear. The manufacturing structure is still immature as garment and textile production and food processing are low value-added and less sophisticated industries. The production of construction materials, electronics, engines and machineries, chemical products, motorbike and car assemblies, plastic products, and other consumption materials is still in the early stage and is used very minimally as import substitution.

As for the labor distribution in the industrial sector, about 50% of the workers are engaged in the garment sector, 25%, in the construction sector, almost 10% in the food/beverage processing, and 13% in other manufacturing segments (Socio–Economic Survey 2007-2012). The organization of labor distribution is comparable to that of the valued-added structure. With regard to the export structure, during the period 2000-2008, the export of textile and footwear and that of wood/wood products accounted for 75.6% and 22%, respectively, of total export. Between 2009 and 2013, however, export of wood and wood products had improved extremely by around 30% per annum, whereas that of textile and footwear dropped to just around 58%. Three main products—agricultural products, rubber, and automotive—have also chiefly contributed to the export performance. In reality, the export of agricultural products, mainly rice, has increased to 4.36% of the total export in 2013. Furthermore, export of light manufactured items, including bicycles, electrical wires, electronic motors, circuits, television parts, toys, furniture, spring and screws and bolts, etc., has increased.

Generally, the manufacturing sector remains rigid and narrow as reflected on the ratio of the number of industrial enterprises over the total number of enterprises, value-added contribution to GDP, and goods for ship. Cambodia's manufacturing sector includes garment and footwear and food/beverage production. The construction sector has seen extraordinary growth. Consequently, for medium-term growth, it is imperative to emphasize the expansion of the production base and differentiation of export products.

3.2 Key Challenges of Industrial Development in Cambodia

Industrial development based on Leadership, Coordination, and Effective Decision Making entails rigorous coordination in each aspect and level, be it at the policy, institutional, regulatory, and managerial levels and in terms of infrastructure development, financial services, and technical and technological capacity along with private sector participation. These components are inextricably intertwined, and their coordination is necessary to offer effective support to solid industrial activities. Additionally, certifying good coordination will help cultivate the investment climate, inspiring more confidence among investors and promoting the reputation of the state as a key investment starting point in the region.

Cambodia lacks strong policies, strategies, and directions for its industrial development. The development of this sector was driven by general market conditions across diverse sectors and geographical locations throughout the state, which give details of the recent situation wherein development appears to be sporadic or very concerted in some sectors or areas but lacks coherence in exploiting investment potential with regard to the provision of support infrastructure, public services, labor supply and so forth. In this respect, the Royal Government should cultivate areas with industrial investment potential by offering adequate public infrastructure investment and other vital public services supporting industrial activities and promoting the development of workers' wellbeing. All of these assistance actions will go a long way in creating an environment conducive for the development and growth of the industrial sector.

As to the regulatory context, comprehensive coordination efforts are necessary with regard to concerns of investment incentives, taxation system, trade regime, transport, and particularly relevant regulatory characteristics. So far, coordination among additional regulators remains problematic, occasionally differing in terms of their own objectives. For instance, the tax system puts excessive weight on tax collection, whereas the trade system pays attention to market openness; the competition framework for its portion remains unfinished, whereas the transport policy lacks comprehensiveness in scope, to mention a few cases. Such absence of coordination gives rise to needless difficulties for the IDP and weakens government efforts to mobilize resources to realize its industrial development goals.

Rigorous investment decision making requires access to strong and transparent information; hence the need for coordination. The provision of poor information causes wastage and means missed investment opportunities. In such circumstances, there is a need to establish a pure mechanism for enabling access to investment information that could decrease searching cost as well as promote broader participation, which in turn will ensure the smooth functioning of more competitive, sustainable industrial development. Broader dissemination of industrial development information also helps foster more creativity in institutional management, design of strategic policies, and advancement of regulatory framework. While a thorough investment and business climate is vital and crucial, the government's various roles in orienting, leading, coordinating, and supporting the private sector is even more critical.

Industrial development necessitates both active leadership and coordination wherein the chief institution plays an energetic professional role in ensuring close cooperation amid the relevant institutions in order to implement the numerous initiatives and policy measures successfully. Indeed, such leadership and coordination must be applied across the board whether in government institutions, private sector entities, and local communities, which can induce their active and effective involvement in fulfilling their own responsibilities. Thus, there is an obligation to strengthen government institutions with pure delineated responsibilities on one hand and to establish detailed manufacturer associations as a mechanism for ensuring joint benefits through data exchange, advocacy, and cooperation on the other, plus disseminating information to an extensive national and international business audience to solicit their support in the industrial development process.

As for the priority industries, the state does not have chief institutions to drive and coordinate related institutions predominantly in initiating intervention, monitoring progress, and offering a troubleshooting solution to address operational concerns. Past experiences have revealed that known institutions have shirked from their responsibilities, further intensifying the problems. As such, coordinating institutions should be able to offer adequate support in the form of technical assistance, training, and investment for some exact activities deemed important, when the responsible institutions have been found to be unsuccessful in their duties. There is a need to have in place a monitoring device to exert extra pressure on the responsible institutions to take measures and solve these problems.

3.2.1 Technical Knowledge and Skills Base

It is essential to make a critical mass and a skill mix of workers, technicians, engineers, and scientists to advance the manufacturing sector development agenda. The restricted pool of trained and skilled workforces, technicians, and engineers will forbid the state from absorbing and utilizing new sciences and technology for industrial development by resorting to hiring expensive expertise from overseas. Cambodia needs to develop the right human resource capacity to maintain its investment policy, the absence of which would extremely affect the competitiveness of its industrial sector.

Generally, skills training remains insufficient to service the industrial sector, which is plagued by low productivity as a result. As most of the Cambodian workers are in the agricultural and service sectors, a shift to the industrial sector would entail a key variation in the labor market, which has a direct impact on employee recruitment techniques, provision of training, and nurturing of work habit and ethics plus expectations on the sector. Teaching workers with low educational attainment is inefficient, demanding a high degree of endurance and tolerance; hence the need to reinforce the official recruitment process coupled with proper training for labors. The goal is to help staff avoid shifting to the informal channel where they are stuck, indebted, and at the mercy of unscrupulous unions. Overall, the system regulating the recruitment process should be revised and amended.

With the high rate of learner drop-out still prevalent, the Cambodian industry is fixed in a work-intensive, low-productivity industry mode. To foster a sound technical base, workers should have finished at least grade 9 for them to possess the basic groundwork to learn technical skills, which is a prerequisite for learning technology. In term of productivity, a low educational attainment among the workers will lead to loss of productivity in the long term as workers are not able to obtain new skills, with no choice but to take on low-paying jobs. In this respect, the government policy must be planned such that opportunities are opened up for workers to continue learning and obtaining new skills.

The morals of technology/science have yet to be mainstreamed in the state's recent social setting. The youth should be inspired and nurtured to like and study science and technology. Establishing science centers that permit them to study and relate to science can arouse their interest and motivate them to make science a routine in solving daily life issues and to hunt for novel ideas to satisfy their future needs. Little attention has been paid to the growth of metrology and standards, which actually deliver the technological backbone to instill end-user belief and certainty in manufacturing production, be they at the level of large factories, enterprises, or handicraft. Moreover, the use of the right metrology and standards adds vast value to the production process through cost reduction, better certainty, and faith in the market in terms of quantity, price, quality, and safety of products. The improvement of metrology and standards is unquestionably key to attractive investors to the industrial sector.

3.2.2 Labor Market and Industrial Relations

Industrial relations are a chief issue for industrial growth not only in the state but also in nearly all nations working over a transition from agriculture/rural to industry/urban situation. The right management of such transition can lay the basis for promising investments in the future, mainly to ensure sound operating conditions, high productivity, and sensible wages for the staff. The process would require vigilant and systematic solutions in accordance with the applicable regulatory agenda so as to reinforce the social investment needed to ease the wage increase burden in order to maintain the competitiveness of the economy. This aspect is even more vital when Cambodia links the ASEAN Economic Community where a wisely managed work mobility can ensure the investment competitiveness of the state. Therefore, sufficient development and effective execution of the regulatory framework are essential. Such regulatory system, together with the right job market management such as employee orientation prior to starting jobs, awareness of the rights and duties of employers/employees, and rational demand for sound employment circumstances are vital and should be extensively executed to ensure the constancy and success of the labor market (IDP, 2015).

4. Science, Technology, and Innovation Policy 2020-2030

4.1 Introduction

Science, Technology, and Innovation (STI) are universally considered the driving force in pursuing sustainable economic development, improving social livelihood, and addressing environmental issues as the three pillars for sustaining mankind's survival and happiness. In the national context, investment in STI with appropriate strategy and policy will enable the government to achieve the national development strategies, leading to the successful realization of Vision 2050 for Cambodia to become an upper middle-income country by 2030 and a high-income country by 2050; in the global context it will help achieve the sustainable development goals (SDGs). STI is of great importance and is indispensable given the fact that it is the knowledge and an integral part of all products, methods, and services as the achievements of mankind. As for STI being an integral part, we need to secure the ability to generate and improve STI for sustainable development. For the knowledge part (invisible or intangible), we need to have strong determination to invest and develop STI concertedly in order to achieve our expected outcomes.

Although STI could be acquired by various means and sources, research and development remain the key task for building and developing technology as well as the ability to absorb and localize technology to meet local needs and to build global competitiveness for domestic products and national outcomes.

Boosting and promoting the three levels of research and development—Basic research, Applied research, and Experimental development—require suitable strategy and methodology based on the context of local evidence. The three main implementing institutions for Research and Development (R&D) include: universities suitable for basic research; research institutes expected to develop new technology; and business enterprises mainly driven by experimental development. The interaction among the three institutions is necessary to ensure that knowledge from research can be transformed into national products and outcomes to meet market demand in a timely manner. Furthermore, introducing new products and outcomes to the market faces challenges, requiring support or intervention through commercialization; a business incubation program is also necessary to attain success from the research results.

Given the limited resources, capacity, and time for development, priority must be given to experimental development through technology adoption and localization. At the same time, national capability to absorb technology will be increasingly strengthened. Realizing this strategic task will bring about economic benefit in the short term, which in turn will contribute to upcoming investment cycles. Investing in basic research and applied research should be gradually increased to achieve higher level of innovation for stronger level of competitiveness and for higher value-added. In promoting and developing STI at this primary stage, the main challenge of the government is the limited resources. Despite this limitation, the government must take the lead in R&D investment for the medium term while pursuing other national prioritized endeavors. This requirement stems from the fact that the private sector lacks appropriate capacity, confidence, and experience in R&D. This, together with the importance of national and international cooperation, requires strong political will of the head of the government in order to succeed in developmental efforts.

Thus, Cambodia must carry out quick intervention through creation of and commitment to implementing an STI policy in order to strengthen STI development and to reduce its gap with countries in the region within the given time and geographical context. Investment in STI is crucial for socio-economic development to achieve the stage-wise realization of national strategies and plans toward Cambodia’s Vision 2050. Naturally, STI development is cross-sectoral and based on highly specialized skills, which in turn requires the concerted efforts of ministries and agencies concerned taking into account the existing roles and responsibilities as well as policies. Given the fact that STI covers a wide range of fields of knowledge, the government needs to set priorities, strengthen and promote human resource based on the education and vocational training system to realize STI outcomes, or transform knowledge as an enabler aiming at achieving the national objectives of competitive production diversification, effective approaches, and attractive services for sustainable socio-economic development. The formulation of national STI policies seeks to establish strategies and approaches with the aim of realizing the predetermined goals in accordance with the actual national context by promoting and leveraging knowledge acquired from education toward a knowledge-based economy, so as to achieve national the development agenda sustainably such as Cambodia's Vision 2050.

4.1.1 Vision

Cambodia is endowed with Science, Technology, and Innovation potential as a driving force for inclusive and sustainable socio-economic development to achieve its Vision 2050.

4.1.2 Objectives

This national policy aims at strengthening the STI foundation, improving STI environment, and developing the STI ecosystem for sustainable development and enhancing the quality of people's life, at all levels and sectors.

4.1.3 Goals

To realize the Vision and Objectives above, based on the conceptual framework of adopting and adapting technologies and further innovating, the following goals are set: i) developing and strengthening adequate STI human resources with quantity, quality, and composition with professional ethics by considering gender equality; ii) enabling prominent STI human resources to perform leading tasks; and establishing filtering and promoting mechanisms in order to create an enabling environment for national STI human resources to realize their full potential; iii) enabling national research and development in an efficient, effective way focusing on the adaptation of the acquired technologies to the local context and enhancing the capacity to absorb foreign technologies; iv) developing and strengthening the dynamic innovation ecosystem with capacity to synthesize technologies and engineering to acquire national outcomes with incremental innovation in order to foster prioritized national industries and businesses for local consumption and export aiming at more productive development; v) instilling in society an STI culture in an inclusive manner, with the aim of securing public confidence and trust on products and services that use national technologies and ensuring that those who made efforts and investments in STI development are satisfied with their outcomes as well as the outcomes of STI governance reform.

4.1.4 Strategies

To achieve the Vision, Objectives, and Goals above and to enable this national STI policy to be implemented smoothly and effectively, the key strategies are set as follows:

- Developing and strengthening adequate STI human resources with quantity, quality, and composition with professional ethics by considering gender equality: a) use, strengthen, and enhance existing human resources through training and refresher training for immediate needs; b) provide opportunities to human resources whose capacity has been built to engage in STI activities and in national and international joint research in order to bridge theory and practice; c) support knowledge and skills transfer from within and outside the country; e) attract talents from abroad with appropriate support and encouragement. f). For the medium and long term, qualify human resources for education and higher education in countries making considerable investments in R&D and STI development. Establishing a network/cluster of intellectuals as well as an industry network within and outside the country based on the trained human resources and with diplomacy support.
- Enabling prominent STI human resources to perform leading tasks and establishing filtering and promoting mechanisms in order to create an enabling environment for national STI human resources to realize their full potential: 1) enable the fostered human resources to perform their duties with full capabilities; 2) monitor, assess, and address problems and challenges of scientist performance; 3) build and operate effectively, fairly, and sustainably the talent assessment mechanism and qualification system with the aim of providing incentives in the form of pecuniary benefits and honorary awards to human

resources delivering outstanding outcomes, in order to promote national talents; 4) create a national STI program for the nationwide youth who are in the capacity development stage.

- Enabling national research and development in an efficient, effective way, focusing on the adaptation of acquired technologies to the local context and enhancing capacity to absorb foreign technologies. a) Increase and manage resources for efficient, effective R&D in response to the particular needs of STI and support the three principles of (1) contest from all institutions and stakeholders and identification of prioritized national projects for outstanding scientists, (2) mandatory existence of STI foundation, and (3) promotion of innovation of products, productivity, and services. b) Secure the actually needed R&D facilities, equipment, and infrastructures. For efficiency of investment, establish prioritized national science research centers for common and specialized uses, operated by trained technicians. These centers are to complement science facilities and infrastructures existing at the various science research institutions. Meanwhile, build an R&D facility and equipment information system with up-to-date data and information on the number, status, and uses of such facilities and equipment. c) Apply a ratio of 1-2-7 for investment in basic research, applied research, and experimental development, respectively, considering the innovation priority, d) Prioritize the necessary technology transfer and import to localize, aiming at incremental innovation based on Make-Some-and-Buy-Some Strategy. e) Set five potential technology domains as national priority: (1) agricultural yield increase, produce diversification, and agro-processing; (2) modern production and engineering; (3) health and biomedical; (4) material science and engineering; and (5) services and digital Economy. Besides, pay attention and make use of emerging technologies such as Artificial Intelligence technology, Space technology, and spatial technology with capability to provide real-time macro management. f) Create a National Research and Development Program to attract and use human resources, financial resources, and national and international cooperation aimed at achieving nationally prioritized STI development.

- Developing and strengthening a dynamic innovation ecosystem with capacity to synthesize technologies and engineering to acquire national outcomes with incremental innovation in order to foster prioritized national industries and businesses for local consumption and export aimed at more productive development: i) create a National Innovation Program to support, promote, facilitate, and bridge higher learning institutions, research institutes, and business enterprises in transforming research into innovation rapidly to benefit society and economy; ii) create a technology transfer office/unit in R&D organizations and institutes, whose duties are accepting and connecting needs in the innovation ecosystem, bridging technology demand and technology development, operating a technology dissemination and demonstration program, and gaining extensive support for innovation under a national coordination mechanism; iii) participate in establishing and supporting public and private entrepreneurship programs providing concessional loan and financial grant, stimulate and promote national and international private-private, public-private, and national-international partnerships, promote and increase industrial participation, and support and increase the capability of small and medium enterprises while stimulating large corporations to achieve higher innovation capability; iv) encourage the building of technology parks and innovation parks as technology and innovation beds to attract and adapt technologies and synthesize engineering for innovation, develop prototype, and incubate national entrepreneurship; v) promote and support the commercialization of products, processes, and services of national technologies and accomplishments in order to gain trust and preference nationally and internationally with national pride; vi) synergize public and private investments in research and innovation to achieve the average ASEAN's Global Innovation Index.
- Instilling in society an STI culture in an inclusive manner, with the aim of securing public confidence and trust on products and services that use national technologies and ensuring that those who made efforts and investments in STI development are satisfied with their outcomes as well as with the outcomes of STI governance reform. 1) Public, private, and society jointly strengthen education, training, and advocating system qualifying "Science for All" with professional

ethics and encourage creativity, innovation, and entrepreneurship. 2) Cultivate trust on national technologies and preference for national products and outcomes by explaining and demonstrating their unique and natural characteristics. Public procurements lead this endeavor by ensuring that procurement items are national value-added or are national outcomes. Meanwhile, the quality and price of the procured items are continuously improved through national research and innovation system. 3) Construct public STI infrastructures for the community such as STI library and museum and arrange STI forums, STI Day, and STI events; encourage participation and support and cultivate creativity and innovation. Increase geographical indication products and entrepreneurship in the community. 4) Offer opportunities for inclusive businesses and social enterprises that fulfill social and environmental responsibilities nationwide.

4.1.5 Action Plan

In order to achieve the strategies of this Policy, the Action Plan is defined as follows:

■ **Development of Mechanism**

To support the implementation of this National Policy, create a mechanism as follows: 1) ministries or institutions concerned establish an office or a unit responsible for STI and have research and innovation activities with the main duties of participating, supporting, strengthening, and bridging own-sector innovation priority and national innovation network; 2) strengthen and facilitate key public STI services such as intellectual property, national technological standard for certification, and technological capability certification in order to increase trust nationally and internationally on national outcomes with national pride and by promoting legacy or existing outcomes; 3) strengthen the roles, responsibilities, and cooperation of national STI organizations at the ministry and national institution level as well as the subnational level, enhance the capability of these organizations in defining and implementing the STI agenda in a concerted manner and in international relations; 4) for the prioritized National Technology Domains and innovation system, strengthen and, if necessary, study and establish research institutes aimed at the localization of national priority technologies to develop product and outcome prototypes as well as study and establish specialized STI agencies to accomplish special missions focusing on innovation; 5) study the possibility of involving Legislative Power in the process of STI implementation in Cambodia; 6) strengthen science diplomacy and have STI agenda in summits to facilitate and implement STI relations across countries smoothly; 7) take part in international STI organizations and jointly support the establishment and strengthening of financial resource network, human resource network, research and innovation network, industry network, and international market network.

■ Development of legal framework

In order to support the implementation process of this National Policy, prepare the legal framework such as law in connection with the basic framework of STI development and other legislations necessary to strengthen governance of promote and stimulate STI development.

■ Financial support

To support the mechanism and implementation of each strategy of this Policy, the sources of financial support include the national budget, which consists of the budgets of the relevant ministry and institution, financial support and grant of Development Partners, contributions of the private sector, community, and individuals, and other sources.

■ Human resource development

To implement this National Policy efficiently, make best effort to build and strengthen sufficient human resource quantity, quality, and composition as follows: a) develop and assign human resource composition for the STI human capital pyramid, which, from top to bottom, consists of (1) the Prime Minister’s STI advisers, (2) STI spokespersons, (3) STI planners, and STI managers, (4) STI educators, and (5) STI practitioners/implementers; b) educate and train researchers and technicians sufficiently for STI activities by each phase of development; c) attract elite knowledgeable and experienced STI dignitaries; d) select local and overseas scientists, professors, and specialists in STI; e) foster government officials who are capable and genius in STI; f) provide education and training to upgrade capability, skills, and ethics of STI human resources.

■ **Implementation process**

The General Secretariat of the National Science and Technology Council is tasked with facilitation; the ministries and institutions concerned are responsible for collaborating on the formulation of a detailed Action Plan relevant to their business domains using their resources, in the short/medium/long term, and with firm commitment to accomplish such.

4.1.6 Monitoring and Evaluation

For the efficient implementation of this National Policy, ministries and institutions in charge have to: a) jointly develop and implement the mechanism and ICT system to monitor implementation and to adjust and evaluate the implementation of this Policy, analyze the implementation result, report on outcomes, challenges, and impacts, and suggest the next step and direction; b) use the results of the evaluation as basis to improve the shortfall and adjust the direction and seek solutions to challenges in order to re-arrange the appropriate plan of activities; c) evaluate the efficiency of the works done and re-orient works under the framework of a centrally coordinating body that is not the direct implementer, which is currently the National Science and Technology Council having delegates from ministries/institutions in the STI management structure.

4.1.7 Conclusion

This National STI Policy is established to support the national development agenda aimed at achieving Cambodia’s Vision 2050 by strengthening the STI foundation and building national STI capability in order to: create potential technologies for development; strengthen innovation capacity in response to the fundamental needs of the nation; improve the quality of people’s life and increase national wealth; develop a competitive national industrial foundation; and improve STI governance. In this regard, the National STI Policy is a tool and a driving force to push forward the socio-economic development to a new phase, sustainably aiming at achieving the national development agenda and not prioritizing well-known science outcomes. Meanwhile, successful STI development requires concerted effort and full-fledged enabling environment at both technical and political levels as stated in this National Policy. In this context, the STI development has to be regarded as prioritized inter-ministerial works to be stipulated in the upcoming national strategies, national policies, and national strategic development plans. In summary, the most important requirement that we need to focus on is to realize the implementation of this National Policy, aiming at the successful implementation of the national development agenda with national pride (STI, 2020).

CHAPTER 4

Strategic Technologies Roadmap Priorities

Capacity Building and Pilot Implementation
Program on Technology Roadmap

Chapter 4. Strategic Technologies Roadmap Priorities

1. Introduction

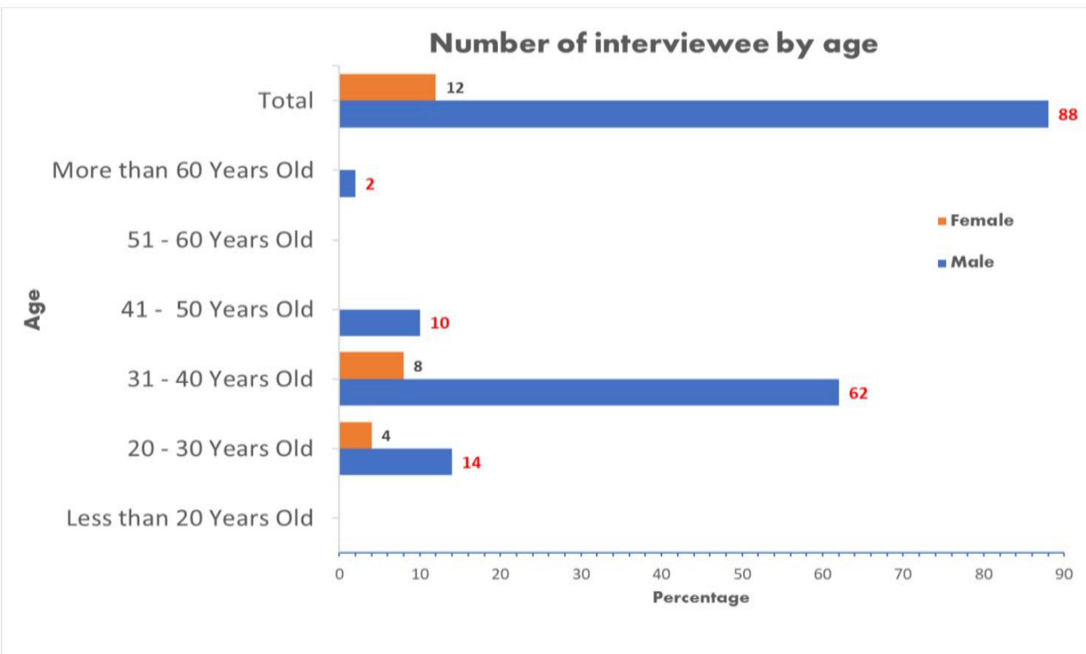
In order to accomplish the government's ultimate goals and strategies as set forth in the Rectangular Strategy Phase IV and the National Strategic Development Plan 2019-2023, RGC does need to carry out the priority development reforms in some potential areas, especially the national strategic technology priority development and support, which are crucial to sustain and keep Cambodia's economy alive amid the effects of the global pandemic. The notable point is the demand-driven technology from both public and private sectors. Cambodia should have preferential policy incentives and tax exemptions for technology equipment and related research equipment in both physical and non-physical aspects from RGC to help reboot Cambodia's economy once again and quickly. Moreover, the government should start thinking about fund injection into R&D, particularly the technology priority approved by the state. It takes years to see the result or to take effect, but it will be a solid foundation for Cambodia's economy and for sustaining growth compared to the neighboring countries that have supported the R&D since their grassroots levels, building the mindset of the locals as to the crucial benefits of S&T development. We acknowledge that there is a limitation in the government's funds for R&D, but there are still many pathways to overcome the current challenges. Together we could eliminate the obstacles, but we cannot do it alone; thus, at the national and sub-national levels, the government and private and relevant stakeholders need to build collaborative linkage to deal with issues and promote S&T development in Cambodia. Most importantly, preparing and implementing the strategic technology roadmap priorities is a vital strategy for realizing a robust Cambodian economy. As we all know, the strategic technology roadmap priorities are a key technology development theme that cannot be achieved without the full participation of all crucial stakeholders and key players related to this area, and this technology development will not replace the technologies that are being implemented by the ministries and relevant entities but is key to helping boost the state's economy. So in this chapter, there will be narration and illustrations about the results of a small survey conducted by the MoP research team investigating the current practice of technology from the public and private institutions that are randomly selected for interview and respondents who filled out the forms. The objective of this small research is to seek the common objective of effective investment from the STI viewpoint from the selected sample size. The detailed purpose of this survey is to see strategic technology candidates and its current activities being conducted in Cambodia and their suggestions; the main technology areas for short term and long term will be highlighted, and the economic impact, technical feasibility, national safety, and security concerns according to the interviewees, especially the factors hindering S&T development and support such as social and ethical acceptability, technological feasibility, industrial and commercial capability, lack of funding, economic viability, and regulatory, policy, and standards of each technology candidates will be identified.

2. Method for Investigation

In order to achieve the objectives above, the team has developed the questionnaires and dataset and organized a list of the top 12 technology areas and some technology candidates to make it easy and convenient for interviewers and respondents in selecting or filling in the 10 technology areas and 10 strategic technology candidates from each question. All experts are cordially asked to answer the questions based on some guidelines and conditions—for instance, policy relation, discrimination, reproducibility, attractiveness, feasibility, and importance are derived to determine the variable for the survey. It has also been highlighted that consensus from all stakeholders on the selected criteria must be obtained before proceeding with the survey, and that the answer for each question as to the most needed or utilized technology by the respective ministries and institutions must be considered. The GS-NSTC staff had been assigned to interview prospect experts from the selected sample size because their thoughts will be key to identifying the current status of technology development and challenges in Cambodia as well as the necessary intervention from the government or authorized ministries. A total of 50 respondents have been interviewed face to face, and the respondents have been carefully selected by considering only those respondents with high educational background related to and who are fully experienced in the field of S&T. We have directed our staff to appoint respondents who have high positions in each targeted group in order to get accurate and valid answers easily based on our questionnaires and secure the confidence of interval and reliability of data. Nonetheless, the small number of respondents for certain technologies will definitely limit the validity of the analysis results.

3. Result of the Survey

[Figure 4-1] Number of Interviewees by Age



The graph above shows the respondents by age group. Initially, the respondents who answered the questionnaire belonged to the 31-40 age group, with men accounting for 62% and women constituting 8%. At the same time, 14% of men and 4% of women who answered the questionnaire belonged to the 20-30 age group. No women belonging to the 41-50 age group answered the questionnaire, whereas about 10% of men who answered the questionnaire belonged to said age group. It can be seen that there are no respondents in the 51-60 age group. Neither were there women interviewees in the age group of more than 60 years old, but about 2% of men who answered the questionnaire belonged to such age group. Lastly, 88% of the respondents who answered the questionnaire prepared by GS-NSTC were mostly men, with women constituting only 12% of the respondents.

[Figure 4-2] Number of Interviewees by Qualification

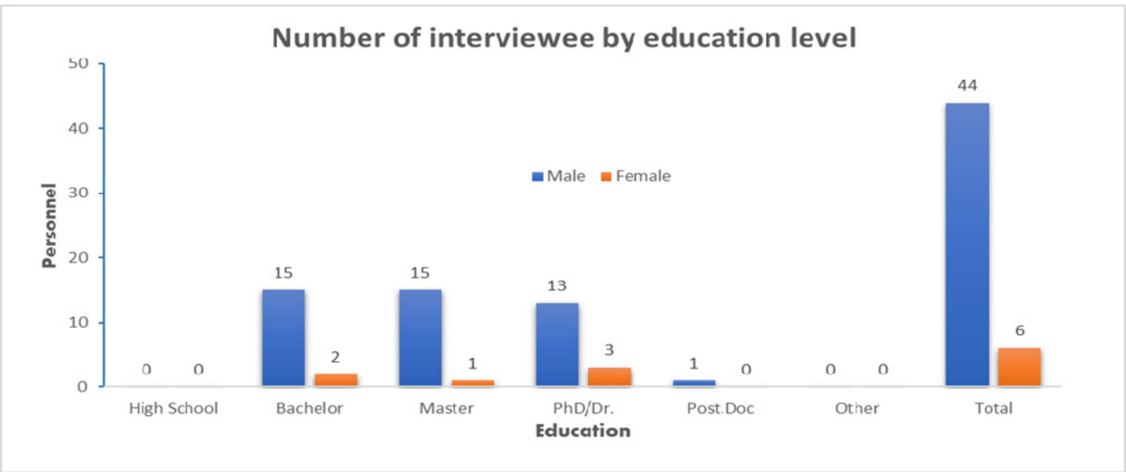
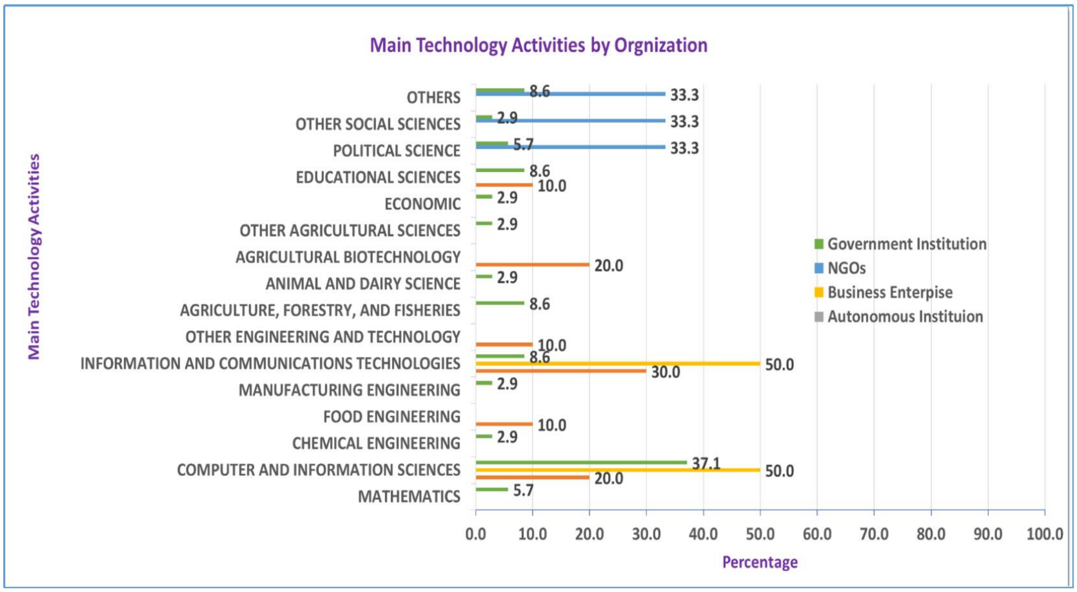


Figure 4-2 above shows that about 4-3 respondents are men with master degrees, with only one female respondent who graduated from university and with one master's degree. The second group of male respondents have bachelor's degrees, with only 2 female respondents having bachelor's degrees. On top of that, about 13 male respondents have PhD/doctorate degrees; surprisingly there were 3 female respondents out of the sample size who have PhD/doctorate degrees. Last but not least, one man with a post-Doc degree was interviewed by the interviewers. In short, 44 men answered the questionnaire prepared by GS-NSTC, with only 6 women taking part in said survey.

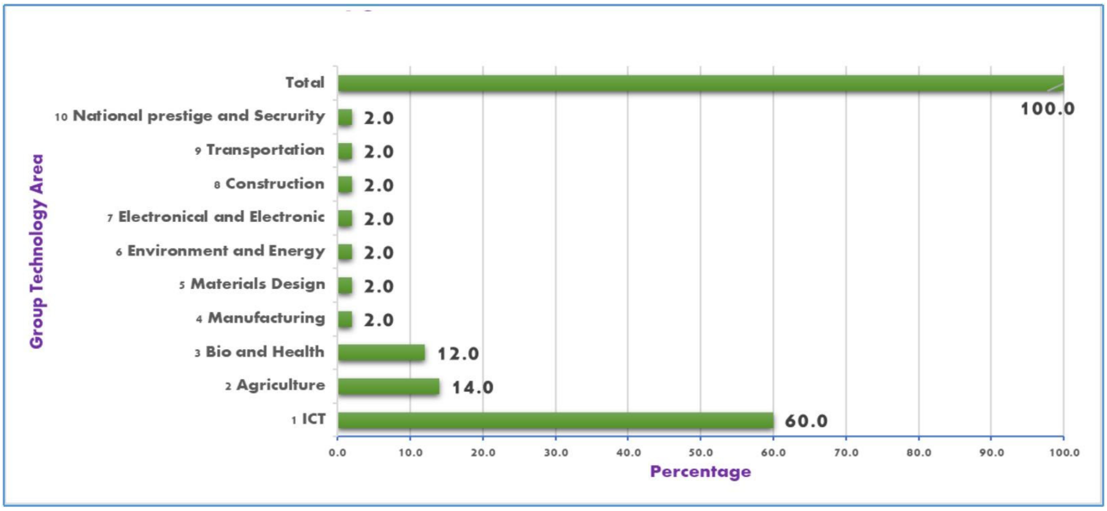
[Figure 4-3] Main Technology Activities by Organization



The image above shows the variety of main technology activities being implemented by the respective agencies or government institutions, including government institutions, NGOs, business enterprises, and autonomous institutions surveyed by GS-NSTC officials. Information and communications technologies and computer and information science constituted about 50% of all activities of the selected targets of this survey. This implies that business enterprises are conducting these activities more actively than government institutions, NGOs, and autonomous institution. Second, there is moderate growth in government institutions with regard to the use of computer and information science in their daily assignments, with 73.1% of the activities by the selected sample size. Meanwhile, there are notable activities being implemented by NGOs related to technologies in other area (33.3%), social sciences (33.3%), and political science (33.3%). On top of that, there are some autonomous institutions conducting technology activities, including information and communications technologies (30%), computer and information sciences (20%), agricultural biotechnology (20%), other engineering and technology (10%), food engineering (10%), and education science (10%). Lastly, government institutions have also conducted a variety of technology activities as shown by the image above, including computer and information science (37.1%), others (8.6%), educational science (8.6%), information and communication and technologies (8.6%), agriculture, forestry, and fishery

(8.6%), mathematics (5.7%), political science (5.7%), chemical engineering (2.9%), manufacturing engineering (2.9%), animal and dairy farming (2.9%), other agriculture science (2.9%), economics (2.9%), and other social science (2.9%).

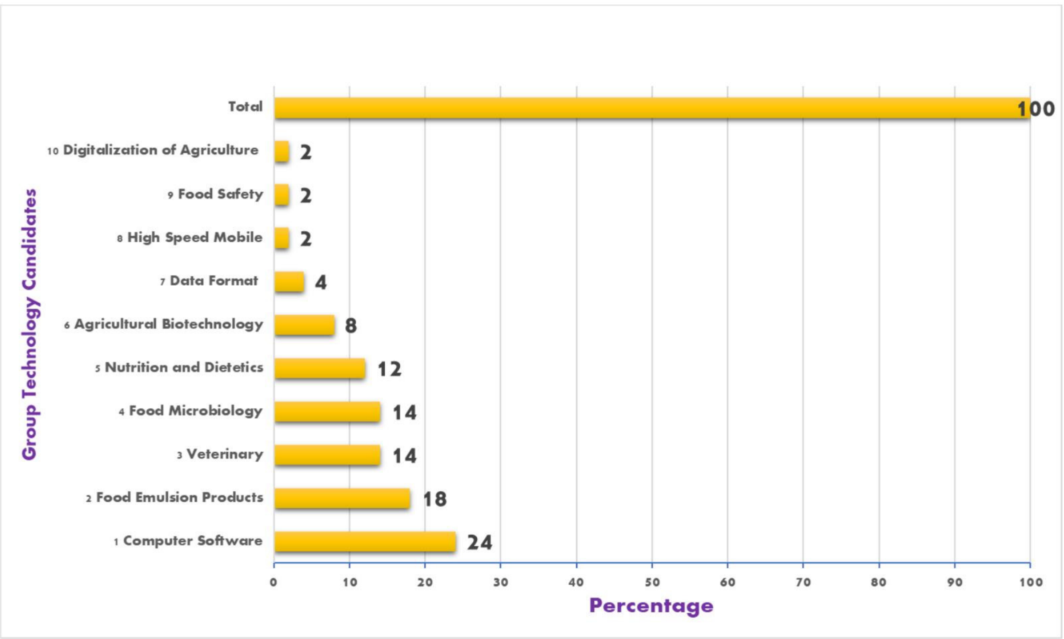
[Figure 4-4] Percentage of Technology Areas Selected by Respondents



3.1 National Strategic Technology for Long-Term

In order to mobilize national resources for the development of science, technology, and innovation and build the principle of core industries in Cambodia, the General Secretariat of the National Science and Technology Council (GS-NSTC), based on majority of the available options, has critically identified the long-term national strategic technologies priorities in ten areas. As could be seen in the image above, ICT, which accounts for 60% of all technologies, is cited by the respondents as the closest to their expertise. It might be true, given the current industrialization era, that Information and communications technology (ICT) is revolutionizing the way people do business and e-government; digital revolution industrial 4.0 is also heard a lot these days. This is characterized by global and digital communications, high-density data storage, and low-cost processing, not to mention an increasing number of global users of digital technologies. It is widely recognized that ICT will be a key driving force in all aspects of development in the next few decades, as it has always been since the information revolution. Second, agriculture technology (14%) was chosen by the interviewees; incidentally, it was the first national technology priority area approved by the National Science and Technology Council of Cambodia. The bio and health technology (12%) area was chosen as the third closest to their expertise. This seems to be contrary to the decision made by NSTC regarding the first national technology area, which is agricultural processing; based on the current trends of technology activities in Cambodia, however, it is just software and digital contents or ICT. The icing on the cake is that there are 8 technology areas that have been chosen by respondents as closest to their expertise, including national prestige and security (2.0%), Transportation (2%), Construction (2.0%), Electronical and Electronic (2%), Environment and Energy (2%), Material Design (2.0%), Manufacturing (2.0%), Bio & Health (2.0%), Agriculture (2.0%), and ICT (2.0%). The respondents answer to the question did not deviate from the main contents outlined in NSDP 2019-2023.

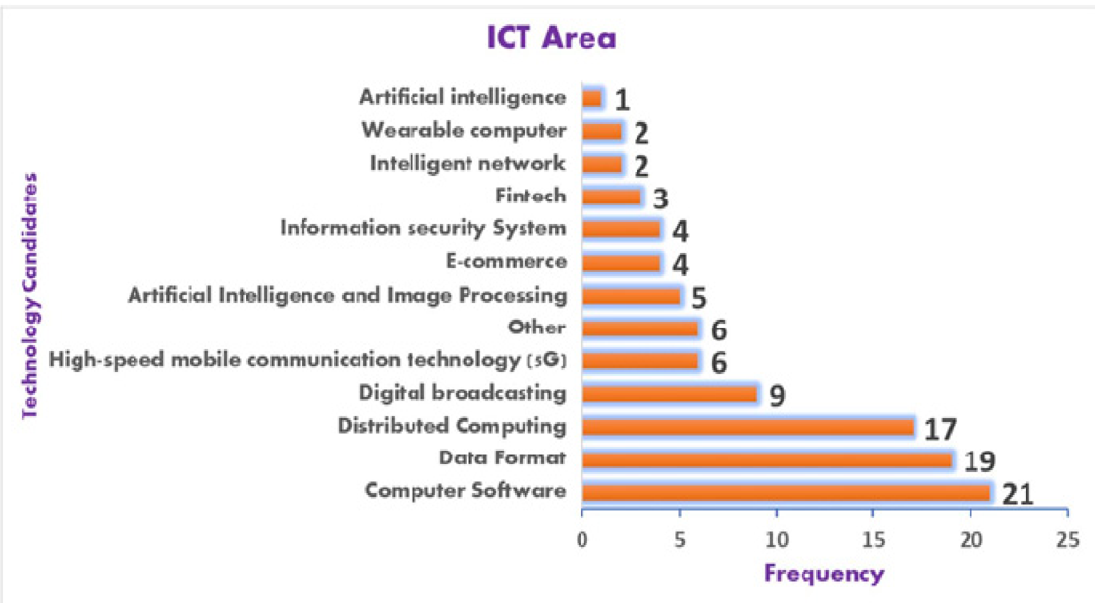
[Figure 4-5] Percentage of Priority Technology Candidates



3.2 National Strategic Technology for the Medium Term

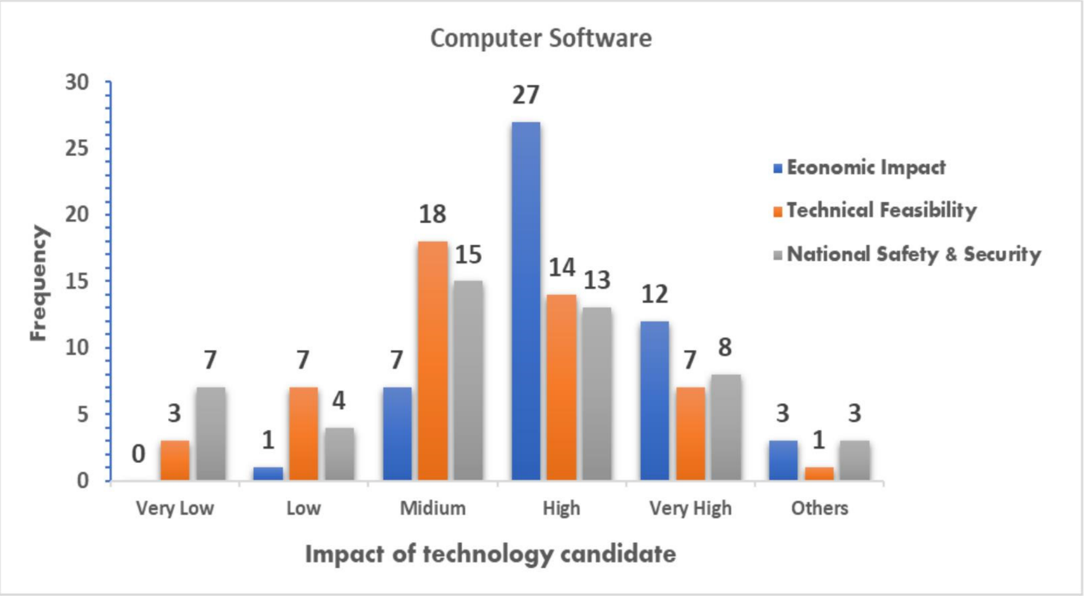
Figure 4-6 shows the proportion of selected priority technology candidates by respondents who perceived that these technology candidates should be planned as national strategy technology. It can be seen that computer software recorded the highest percentage, about 24%, among the 10 priority technology candidates. Meanwhile, food emulsion products (18%), veterinary (14%), food microbiology (14%), and nutrition and dietetics (12%) have very nearly the same percentage. Similarly, agriculture biotechnology accounts for 8%. In short, data format (4%) has 2 times higher percentage compared to high-speed mobile (2%), food safety (2%), and digitalization of agriculture (2%).

[Figure 4-6] Information and Communication Technology Area



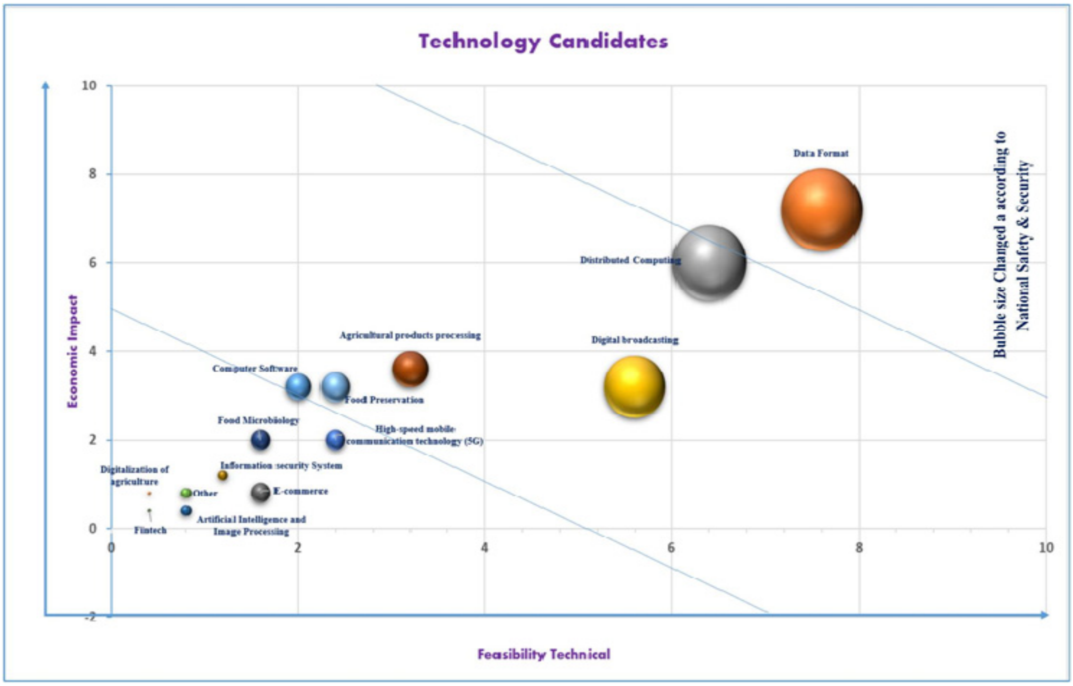
The figure above shows the variety of technology candidates in the technology priority area called the ICT area. It can be seen that the most popular technology activity being implemented in Cambodia is computer software, chosen by 21 respondents among the technology candidates. On the other hand, data format (19) is similarly the second most popular activity being conducted in Cambodia by targeted individuals. Correspondingly, distributed computing (17) is likewise more productive in its technology candidate areas by Cambodia's society. It can be understood that digital broadcasting (9) is also popular and is being utilized by society and both public and private sectors. Meanwhile, there are several other well-known technology candidates in this technology area known as ICT that are being implemented by respondents, including artificial intelligence and image processing (5), e-commerce (4), information security system (4), fintech (3), intelligent network (2), wearable computer (2), and artificial intelligence (1). All these technology candidates are quite popular in Cambodian society. Still, it seems to be contrary to the national technology priority approved by the council. With the lowest percentage assigned to artificial intelligence—chosen by only 1 respondent— it is not easy to cultivate the next generation of Cambodian youngsters who will be able to learn and have the capacity to learn coding because this technology candidate requires the individual to have strong background in STEM education and demands lots of efforts, support, and funding for research. Nonetheless, the current challenge is that there are no clear policy and incentive mechanisms for boosting and promoting S&T talents both in Cambodia and abroad.

[Figure 4-7] Impact of Technology Candidate



Based on the response from the interviewees, we can definitely see that computer software as a top technology candidate has been selected by the respondents as one having high economic impact (27) among the 50 targeted respondents compared with technical feasibility (14) and national safety and security (13), which were also perceived as high-impact. Meanwhile, 18 respondents answered that computer software has moderate technical feasibility, 15 interviewees responded that it is at a moderate level in terms of national safety and security, and only 7 respondents acknowledged that it has moderate economic impact. Two respondents believe that computer software has very high economic impact on Cambodia's economy, and even the government is promoting STEM education to catch up with industry 4.0 and digital economy, especially the techno-entrepreneurs in the country. On top of that, eight interviewees answered that computer software will have very high impact on national safety and security in terms of preventing cyber-attack and related Internet crimes by skillful hackers. There are several cases of unknown hackers hacking the government website and stealing vital data and causing server downtime. This nation is not yet ready to transform into e-government or even for e-commerce because there is a limitation in terms of talented skills and resource persons as well as the right support from the government in controlling and managing the servers; most importantly, the local market depends on outsiders to monitor and prevent hacking by various means; thus, subscribers need to pay to secure their data; in particular, the cost of maintaining the servers are passed to them due to the shortage of skillful government officials and in some private sectors. Most banks in Cambodia pay outsiders to ensure the smooth operation of the bank's servers, and annual spending for maintenance has been quite hefty.

[Figure 4-8] Assessments of All Technology Candidates



This is how we assess the technology candidates from the pool of technology candidates selected by our respondents as shown in the bubble analysis by comparing in terms of economic impact, technical feasibility, and national safety and security. Initially, we need to use quantitative assessment and weight analysis aimed at picking a pool of national strategic technologies from the initial list of national strategic technologies. Once the parameters are compiled, and each technology is weighed, all technology candidates will be represented graphically in a two-dimensional graph ranking the individual technologies, and the results of the most important technology candidates will be displayed in bubble chart format as can be seen in Figure 4-8 above. The bigger bubble indicates the biggest number of occurrences based on the vertical and horizontal axis of each variable or parameter. Simply put, according to Figure 4-8, the organization of information according to preset specifications (usually for computer processing), data formatting, format, computer formatting, computing device, computing machine, data processor, electronic computer, and information processing system—a machine for performing calculations automatically—is evaluated as the most important technology and has been considered one of the highest rated according to scoring, scaling, and weighing assessments among the three parameters such as economic impact, technical feasibility, and national safety and security, representing the biggest size of the bubble plots. Still, technological development cannot rely on the ICT component alone. Agricultural products and processing and biotechnology in food products should also be paid more attention considering the results of weighing assessments of economic impact, technical feasibility, and national safety and security. Notably, for computer software was chosen by the biggest number of respondents, but their ratings were lower than previous technologies because of the weighing and scaling assessments between the three parameters or variables, which analyzed economic impact, technical feasibility, and national safety and security, compared with data format, distributed computing, digital broadcasting, and agricultural products processing.

[Figure 4-9] Factors Hindering Technology Development

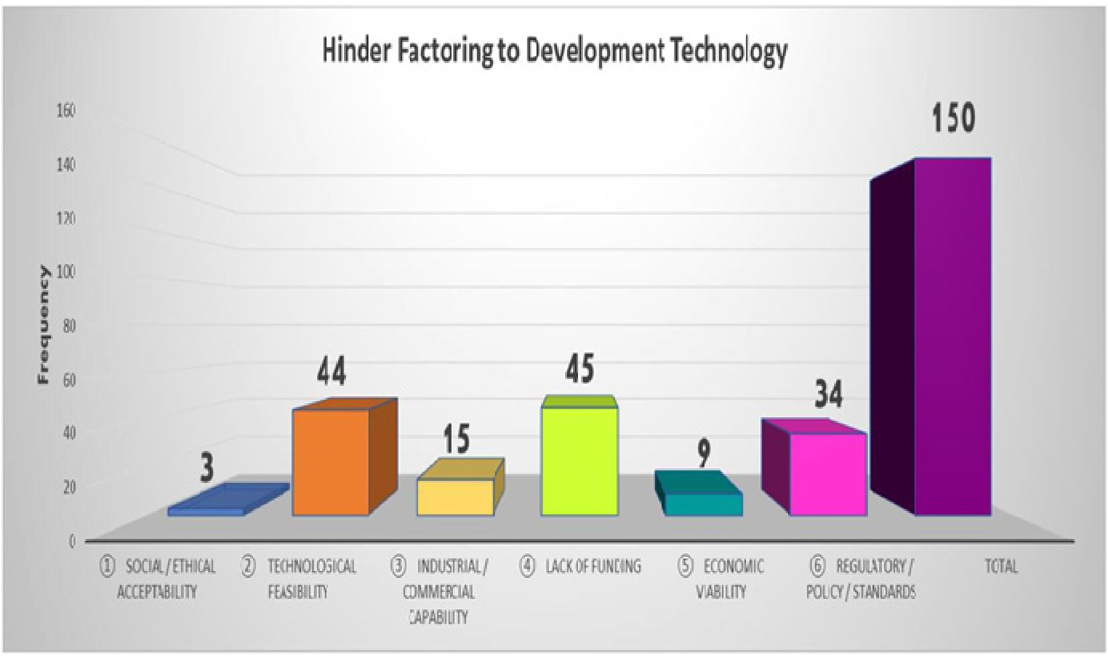


Figure 4-9 shows that there are several factors affecting technology development in Cambodia according to the answers of our respondents. It can be seen that the lack of funding (45) and technical feasibility (44) are the most important factors affecting the technology development of Cambodia. Without funding system or funds, there will no technology activities because researchers or practitioners and relevant stakeholders need money to function and research equipment to conduct their research projects or test the ideas. Policy and standards (34) to support them are also critical because, without clear policy support, protection of rights, and standards, they can hardly operate or carry out their technology activities or development. Thus, affirmative actions from the government with regard to policy and regulatory supports and standards are a must and should be considered by the government. Only technology development and innovation will help boost local business in Cambodia.

Nonetheless, there are other factors affecting technology development in the state, like obstacles in finding industrial or commercial capability (15) to sell their final products. There are many concerns from the ICT producers' areas and related areas with regard to the difficulty in finding the right market locally and abroad to sell their final products as Cambodia's market in terms of technology is still narrow. The government needs to join hands in overcoming the current challenges being faced by the local market. Moreover, economic viability as cited by 9 respondents means that there are lot of challenges toward technology development without thinking clearly about the cost-benefit management and with less budget to conduct the technology development. Some activities are being implemented based on donors' budgets that are allocated late or which take time to receive. Thus, developers will likely not think carefully about economic viability, and research and technology development will eventually fail or prove to be unsuccessful. Additionally, other findings reveal that the biggest challenges to adopting the use of new technologies are hardware incompatibility, complexity, language barriers, lack of electricity, computers, Internet access, and practice for trainees as well as the inability to understand the advantages of these technologies.

To conclude, there are a lot of important technology areas, along with the strategic technology candidates that have been narrated from the technology development perspectives and the technology policy preparation for long- and short-term strategy development in Cambodia. These results will serve as the most important document in assisting and supporting the right decision making and intended users who demand understanding the current technology area and strategic technology candidate's status and development in this nation. Information and communication technology is the most popular area of technology development in Cambodia. It is only right to continue supporting ICT because it has limitless benefits that can support Cambodia's economy. Currently, we have several ministries—such as the Ministry of Industry, Science, Technology, and Innovation, Ministry of Posts and Telecommunications, Ministry of Commerce, and Ministry of Labor and Vocational Training—National Bank of Cambodia, and Ministry of Education, Youth, and Sport utilizing ICT, but the government needs to take into account transforming the agriculture sector into smart agriculture and digital agriculture, doing away with conventional or traditional practices, especially to harness the optimal benefits from industrial revolution 4.0. By the way, there are several barriers in implementing ICT due to lack of skilled personnel to manage and function in the respective ministries; for instance, data format is considered a tough job to manage because it requires skillful personnel to prepare and process it effectively and conveniently. Even the private sector lacks knowledgeable staff to perform the work because it demands individuals with high educational background and experience, especially those who are good at mathematics. Data format is highly valued and a big concern for national safety and security to prevent both black and white hackers as well as distributed computing and digital broadcasting. On top of that, computer software, which was chosen by the biggest number of respondents, is operated by the private sector rather than government institutions or entities. Several challenges have been raised by software developers such as marginal market share or narrow market to sell the products generated by the private company. Moreover, Cambodia does not have any effective support mechanisms or funds to help operate and even start a business. Sometimes, they have talented programmers and software developers to produce final products that are ready for selling, but they could not export them. In this connection, from the state perspective, computer software is extremely limited, we have difficulty finding skillful programmers or software developers working for government services, but they could be found in the private sector. It could be a matter of salary or entrance exam for government officials.

Moreover, the government needs to invest in or provide preferential benefits or incentives for bio-research skills, experiences, and related equipment to the most talented personnel or universities with the right skills to start doing research in the bio-health sector. Agriculture and bio-health should integrate and develop together as the country has an abundance of natural resources to process and convert into final products instead of importing from neighboring countries, but we have a limitation of support including research equipment, funds, and effective mechanisms from the state to do research in those potential areas. The Ministry of Agriculture, Forestry, and Fisheries is a leading ministry in this area; from the research perspective, however, this ministry is still limited because of the allocated budgets and support from the leaders. Especially research equipment usually costs a lot, and spending on regular maintenance and repairs is inevitable. Most importantly, we should avoid importing what we can produce and start investing in the state to promote the local business environment. As we understand, there are several factors affecting technology development in the state among practitioners or researchers and all relevant stakeholders, and we believe the government needs to step in and ease the burdens of those individuals. Clear budget allocations and initiatives from the government are a must to support them in developing and implementing their plans or technology activities. Sometimes, they have great ideas that lack technology feasibility; thus, the relevant ministry with the most required skills can offer training or coaching. As such, strategic technology priorities are the main strategic technology development to help pave the pathway and to sustain Cambodia's economy growth and development in the right way.

CHAPTER 5

Conclusion

Chapter 5. Conclusion

1. Implications of the Research

Cambodia is working on various national strategies and plans for national development, such as the Rectangular Strategy Phase IV and National Strategic Development Plan 2019-2023. To achieve the goals of these strategies and plans, the development of national strategic technologies is essential. Since the development of science and technology is a key factor in economic development, the government should establish a foundation through policy and budget input for the development of the science and technology field from a long-term perspective. In particular, the selection of strategic technologies and the establishment of roadmaps are key issues for solid national economic development, so they should be done with the sufficient participation of experts in each field. In response, the Cambodian Planning Department sought to identify candidate strategic technologies on a small scale and examine their economic impact, technical feasibility, national safety and security impacts, industrial capabilities, and social acceptability.

As for the implications of the series of activities, first, Cambodia’s national strategic technology, which is essential for the establishment of a national strategic technology roadmap starting next year, was selected. As a result of the exchange and education and training conducted through the project last year, it was confirmed that GS-NSTC could lead the development of national strategic technologies for its experts. In addition, data on Cambodia's national strategy on science and technology were collected in the process of selecting the National Strategic Technology Army.

Second, they have accumulated experience in carrying out projects that can cope with unpredictable external factors that may arise in ODA projects, such as COVID-19. Full-scale start of business was delayed compared to normal, with situations that have never been experienced before and are difficult to predict even a month later. With direct exchange expected to be difficult next year due to COVID-19, this experience has enabled us to map out plans for next year's projects preemptively. In addition, institutional manuals and inter-agency ties have been established to maintain cooperation even when such situations occur.

Through this experience, two implications were drawn. First, the selection of strategic technology requires a process that reflects the characteristics of the country. Most developing countries are interested in the hi-tech industries and technologies of advanced countries. As Cambodian experts' interviews with the computer program show, however, it is important to select strategic technologies considering the economy, security, feasibility, and national development goals of the country. Second, it is worth considering the implementation of the service contract method for the partner agencies of the country in carrying out the project in the future. Due to the difficulty of exchanging research teams between the two countries due to the COVID-19 crisis, the project requested a service survey to a cooperative organization of the Suwon bureau, which had the following advantages and disadvantages. The advantage is that it has served as an opportunity for partner agencies in Suwon to participate in the project more actively with responsibility, and that they can provide learning opportunities by carrying out policy activities on their own. On the other hand, there was no opportunity for Korean research teams to participate in expert meetings or research activities in the region and to think about them together and consult them through face-to-face meetings. Even if the COVID-19 situation is stable, it is also desirable to place parallel orders for some service projects as a way to encourage the active participation of partner agencies in carrying out policy consultations.

Through this survey, the project aims to establish business plans for 2021 and 2022. The main goal is to build three technology roadmaps in 2021 and three in 2022 for six national strategic technologies. Next year, as the project will be changed to a convergence project, a consortium will be formed with South Korean institutions specializing in Cambodia's strategic technology sector. As there has long been a demand from Cambodian experts to cooperate with Korean experts, we hope to create an environment where technical and Cambodian experts can work together. In addition, as direct exchange becomes difficult due to the COVID-19 situation, and the direction of the project will be diversified due to the formation of consortia, we will establish a system that can effectively build and manage the project.

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Appendix 3. Questionnaires of National Strategic Technologies Roadmap Survey

Royal Government of Cambodia

Ministry of Planning

CONFIDENTIALLITY OF DATA

Information collected in this survey is strictly confidential and will be used for statistical purposes only without revealing personals or institutional data.

QUESTIONNAIRE OF NATIONAL STRATEGIC TECHNOLOGIES RODAMAP SURVEY 2020

General Secretariat of the National Science and Technology Council

(OBJECTIVE OF SURVEY)៖

This questionnaire is designed to identify the national strategic technologies for Cambodia for the next ten years.

This survey is being conducted to help fulfill the planning needs of the National Science strategic technologies for Cambodia Program as to technology assessments of great value to the nation.

Which are considered to be important for the future of Cambodia. By definition, a priority is something that receives focus and that enjoys special preferential in terms of financial resources and other forms of incentives.

Definition:

- **Technology:** refers to the state of knowledge concerning ways of converting resources into outputs.
- Source Publication: OECD Productivity Manual: A Guide to the Measurement of Industry-Level and Aggregate Productivity Growth, OECD, Paris, March 2001, Annex 1 – Glossary.

(INSTRUCTIONS) ៖

The questionnaire should be completed by the top manager(s) responsible for business operation or financial operation/management or research and development or corporate planning. Please respond to all questions, unless otherwise instructed.

SECTION 1: IDENTIFICATION

A. Survey area information		Code	
Municipal, Province			
City, District, Khan			
Commune, Sangkat			
Village, Mundol			

B. Respondent Information							
Respondent's name							
Position							
Gender		Age		Occupation fields		Educational Background	
Male	☐	(Below) 20	☐	Business	☐	High school	☐
Female	☐	21~30	☐	Education	☐	Bachelor	☐
		31~40	☐	Research (public/private research institutes)	☐	Master	☐
		41~50	☐	Government Officer	☐	PhD/Dr.	☐
						Post.Doc	☐
		51~60	☐	Other, please specify:	☐	Other, please specify:	☐
		(Above) 60	☐				
Length of Work in your Occupation							
Telephone Number							
Email Address							
Unit-Institution Address							

C. Enumerator Information	
Enumerator's Name	
Signature	Tel.
Interviewing date	DD-MM-YY
Delivery date	Collecting data

Supervisor's Name	Signature	DD-MM-YY

D. Unit-Institution's Profile	
Name of Unit-Institution (Both languages)	

Type of Unit-Institution	Please tick accordingly
1. (Government Unit-Institution)	☐
2. (Higher Education)	☐
3. (Autonomous Unit-Institution)	☐
4. (Business Enterprise-Corporative)	☐
5. (NGOs)	☐
(Other, please specify) …………………………………………………	☐

SECTION 2: ACTIVITYOF UNIT-INSTITUTION

2.1. (Main Technology Activity)

Please describe your main technology activities

……

……

(Code) ☐☐☐ (GS-NSTC Staffs Only)

2.2. **(Secondary Technology Activity)**

Please describe your secondary technology activity

……

……

(Code) ☐☐☐ (GS-NSTC Staffs Only)

SECTION 3: NATIONAL STRATEGIC TECHNOLOGY FOR LONG-TERM

3.1. (Main Technology Priorities)

Q3.1 Select one technology area is closest to your expertise from the technology areas below.

Technology Area	Please tick accordingly
Information/Communication Technology	☐
Biotechnology	☐
Agricultural Technology	☐
Medical and Health Technology	☐
Environmental and Energy Technology	☐
Construction Technology	☐
Manufacturing Technology	☐
Materials Technology	☐
Transportation Technology	☐
National prestige & security	☐
Tourism	☐
Art (Arts, History Of Arts, Performing Arts, Music)	☐

SECTION 3: NATIONAL STRATEGIC TECHNOLOGY FOR LONG-TERM

3.1. (Main Technology Priorities)

Q3.2 Please select strategic technology candidates closest to your expertise in the technology area at least 3 (that you selected in Q3.1) For example, below:

Technology Area	Strategic Technology Candidates	Please tick accordingly
Information/Communication Technology	Artificial Intelligence and Image Processing	☐
	Computer Software	☐
	Data Format	☐
	Distributed Computing	☐
	High-speed mobile communication technology (5G)	☐
	Mobile multimedia contents	☐
	Intelligent network	☐
	Wearable computer	☐
	Digital broadcasting	☐
	E-commerce	☐
	Fintech	☐
	Information security System	☐
	Intelligent robot	☐
	Artificial intelligence	☐
	Home networking	☐
	Intelligent medical system	☐
	If any…...	☐
		☐
		☐
Biotechnology	Additives and Contaminants	☐
	Abiotic Stress	☐

	Food Emulsion Products	☐
	Food Microbiology	☐
	Food Preservation	☐
	Food Quality	☐
	Food Safety	☐
	Food Security	☐
	Food Toxicology	☐
	Functional Food	☐
	Improvement of Finished Products	☐
	New Food Ingredients	☐
	Nutrient Bio-ability	☐
	Nutrient Composition of Food	☐
	Biofuels	☐
	Industrial Biotechnology	☐
	Nutrient Supplementation.	☐
	If any…...	☐
		☐
		☐
		☐
	Animal and Dairy Science (Production)	☐
	Horticultural Production (Husbandry)	☐
	Pest and Disease Management	☐
	Veterinary	☐
	Agricultural Biotechnology	☐
	Agricultural machinery	☐

Agricultural Technology		Seeds development	☐
	Fertilizer technology		☐
	Agricultural water management		☐
	Agricultural products processing		☐
	Digitalization of agriculture		☐
	Irrigation		☐
	If any…...		☐
			☐
Medical and Health Technology	Basic Medicine		☐
	Clinical Medicine		☐
	Medical Biotechnology		☐
	Medical Microbiology		☐
	Medical Physiology		☐
	Medical Diagnostic Imaging		☐
	Medical Biochemistry and Clinical Chemistry		☐
	Human Movement and Sports Science		☐
	Nutrition and Dietetics		☐
	Pediatrics and Reproductive Medicine		☐
	Biomedical		☐
	New Drug discovery and development (chemical)		☐
	New Drug discovery and development (natural)		☐
	Disease diagnostics, treatment, and prevention		☐
	If any…...		☐

		☐
		☐
		☐
Environmental and Energy Technology	Reduction of environmental pollution	☐
	Recycling system	☐
	Natural disaster prediction and damage reduction	☐
	Energy saving	☐
	Renewable energy	☐
	Fuel cell	☐
	Hydropower technology	☐
	Geothermal energy technology	☐
	Wind energy technology	☐
	Solar PV energy technology	☐
	Nora technology	☐
	If any…...	☐
		☐
		☐
		☐
Construction Technology	Prefabricated building	☐
	Concrete additives	☐
	Mixing and pouring	☐
	Green building	☐
	Earth construction	☐
	Raw materials crushing	☐
	Cement Grinding	☐

	Pyro -processing	☐
	Bamboo construction	☐
	Stone construction	☐
	If any…...	☐
		☐
		☐
		☐
Manufacturing Technology	Smart Factory	☐
	Machinery	☐
	New material development (metal, ceramic, polymer)	☐
	Textile Technology	☐
	Automotive Mechatronics	☐
	Manufacturing Processes and Technologies (Excl. Textiles)	☐
	Manufacturing Robotics and Mechatronics (Excl. Automotive Mechatronics)	☐
	Visualization, Information, and Digital	☐
	Advanced, Sensing, Control and Platforms	☐
	Advanced, Materials, Manufacturing Critical, Materials Reprocessing	☐
	If any…...	☐
		☐
		☐
		☐
	Ceramics	☐
	Composite and Hybrid Materials	☐
	Compound Semiconductors	☐

Materials Technology	Elemental Semiconductors	☐
	Functional Materials	☐
	Glass	☐
	Metals and Alloy Materials	☐
	Organic Semiconductors	☐
	Polymers and Plastics	☐
	Timber, Pulp and Paper	☐
	Nanomaterials	☐
	If any…...	☐
		☐
		☐
		☐
Transportation Technology	Self-Driving Automobiles	☐
	Smart Cars	☐
	Next-Gen GPS Devices	☐
	High-Speed Rail Networks	☐
	Gyroscopic Vehicles	☐
	Automatic train coupler	☐
	Roller coaster technology	☐
	Tram technology	☐
	Solar-powered roadways	☐
	Smart pavement	☐
	Glow in the dark roads	☐
	Electric priority lane for charging electric vehicles	☐
	Traffic detection	☐

	tackling natural disasters	☐
	If any…...	☐
		☐
		☐
National prestige & security	Aerospace technology	☐
	Food security	☐
	Air Force	☐
	Army	☐
	Command, Control, and Communications	☐
	Emerging Defense Technologies	☐
	Intelligence	☐
	Logistics	☐
	National Security	☐
	Navy	☐
	Personnel	☐
	Isometric Authentication.	☐
	Defense Communication System	☐
	For spying	☐
	If any…...	☐
		☐
		☐
Tourism	Impacts of Tourism	☐
	Tourism Forecasting	☐

	Tourism Management	☐
	Tourism Marketing	☐
	Tourism Resource Appraisal	☐
	Tourist Behavior and Visitor Experience	☐
	Tourism not elsewhere classified	☐
	Impacts of Tourism	☐
	Tourism Forecasting	☐
	Tourism Management	☐
	Tourism Marketing	☐
	Tourism Resource Appraisal	☐
	If any…...	☐
Art (Arts, History of Arts, Performing Arts, Music)	Art Criticism	☐
	Art History	☐
	Art Theory	☐
	Visual Cultures	☐
	Creative Writing (incl. Playwriting)	☐
	Dance	☐
	Drama, Theatre and Performance Studies	☐
	Traditional Performing Arts	☐
	Music Composition	☐
	Music Performance	☐

	Music Therapy	☐
	Musicology and Ethnomusicology	☐
	Pacific Peoples Performing Arts	☐
	Crafts	☐
	Fine Arts (incl. Sculpture and Painting)	☐
	Lens-based Practice	☐
	Performance and Installation Art	☐
	Cinema Studies	☐
	Computer Gaming and Animation	☐
	Electronic Media Art	☐
	Film and Television	☐
	Interactive Media	☐
	Journalism Studies	☐
	Professional Writing	☐
	Technical Writing	☐
	Journalism and Professional Writing not elsewhere classified	☐
	Cinema Studies	☐
	If any…...	☐
		☐
		☐
		☐

SECTION 3: NATIONAL STRATEGIC TECHNOLOGY FOR LONG-TERM

3.1. (Main Technology Priorities)

Q3.3 Please select all main strategic technology candidates closest to your expertise at least 3 in the technology area (that you selected in Q3.2).

Strategic Technology Candidates	Please description
1. …………………………………………………………………………	
2. …………………………………………………………………………	
3. …………………………………………………………………………	
4. …………………………………………………………………………	
5. …………………………………………………………………………	
6. …………………………………………………………………………	
7. …………………………………………………………………………	
8. …………………………………………………………………………	
9. …………………………………………………………………………	
10. …………………………………………………………………………	

SECTION 3: NATIONAL STRATEGIC TECHNOLOGY FOR LONG-TERM

Q3.4 Respondents are requested to rate the strategic technology candidates (that you selected) according to the criteria and based on the following rating: (5-point scale)

1. (Very low) 2. (Low) 3.(Middle) 4. (High) 5.(Very high)

Strategic Technology Candidates (that you selected)	Economic Impact	Technical Feasibility	National Safety & Security
1. ……………………………………………………………… ………………………….			
2. ……………………………………………………………… ……………………			
3. ……………………………………………………………… ……………………			
4. ……………………………………………………………… …………………			
5. ……………………………………………………………… ……………………			
6. ……………………………………………………………… ……………………			
7. ……………………………………………………………… ……………………			

SECTION 4: HINDER FACTOR

Q4.1 (Which factors hinder the development of this technology candidate? Select three most important factors from the list in order of significance.)

Hinder Factors	Please tick accordingly
① (Social / ethical acceptability)	☐
② (Technological feasibility)	☐
③ (Industrial / commercial capability)	☐
④ (Lack of funding)	☐
⑤ (Economic viability)	☐
⑥ (Regulatory / policy / standards)	☐